Natural Capital Accounts (NCAs) in Eastern and Southern Africa

Diagnostic Review

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1 Introduction

The System of Environmental Economic Accounting (SEEA) (European Commission et al., 2013) is the international statistical standard for measuring the environment and its relationship with the economy. It covers 36 accounts, which are detailed in Table 1.

This work aims to review the NCA data produced for Eastern and Southern Africa and prepare relevant applications that can inform current World Bank priorities and the usability of the data for planning and policy. We conducted a review and comparative analysis of the most recent natural capital accounts (from 2015 onward) prepared by countries in Eastern and Southern Africa, including the Democratic Republic of Congo, Ethiopia, Kenya, Madagascar, Mauritius, Rwanda, Uganda, South Africa, and Zambia.

2 Regional Assessment

The 2025 Global Assessment of SEEA Implementation, which documents the state of natural capital accounting across all UN member states, was complemented with an individual review of countries that have made efforts towards Natural Capital Analysis in any form, regardless of SEEA implementation status. This review draws on the Africa and Far East (AFE) group, covering 26 countries in Eastern and Southern Africa. For each country, an evaluation was conducted whether 36 distinct SEEA accounts, spanning the SEEA Central Framework (CF), SEEA Ecosystem Accounts (EA), and a set of Thematic Accounts, have been compiled at least once.

Accounts are grouped into three broad categories: the SEEA CF covers material and monetary flow and asset accounts; the SEEA EA covers ecosystem extent, condition, services, and monetary asset accounts; and Thematic Accounts cover specialised topics such as forests, ocean, and protected areas.

2.1 Implementation Status

Of the 26 AFE countries assessed, 11 are actively implementing the SEEA in some form, while the remaining 15 have not yet begun. Among those implementing, countries are classified by their stage of progress under the SDG 15.9.1b indicator, *"Integration of biodiversity into national accounting and reporting systems, defined as implementation of the System of Environmental-Economic Accounting"* (European Commission et al., 2013). Compilation refers to the production of a single account, while dissemination refers to the publication of that account. Regular compilation and dissemination refers to the production and publication of an account on a regular basis. Stage I countries have compiled at least one SEEA CF account; Stage II countries have additionally compiled at least one SEEA EA or Thematic account; and Stage III countries demonstrate a sustained institutional programme of accounting across

Table 1: Accounts of the System of Environmental and Economic Accounts (SEEA)

Category	**Account**
Material flow	Material flow accounts
	Economy wide material flow accounts
Water	Physical supply and use tables for water
	Monetary supply and use tables for water
	Physical asset accounts for water
	Water emission accounts
Energy	Physical supply and use tables for energy
	Monetary supply and use tables for energy
	Physical asset accounts for mineral and energy resources
	Monetary asset accounts for mineral and energy resources
Air emissions	Air emission accounts
Land	Physical asset accounts for land
	Monetary asset accounts for land
Agriculture forestry and fisheries	Physical asset accounts for timber resources
	Monetary asset accounts for timber resources
	Physical asset accounts for aquatic resources
	Monetary asset accounts for aquatic resources
	Asset accounts for other biological resources
	Other agriculture, forestry and fisheries accounts
Waste	Waste accounts
Environmental protection and management expenditure	Environmental protection expenditure accounts
	Resource management expenditure accounts
	Environmental goods and services accounts
Taxes and subsidies	Environmental taxes accounts
	Environmental subsidies accounts
Ecosystem extent	Ecosystem extent accounts
Ecosystem condition	Ecosystem condition accounts
Ecosystem services	Physical ecosystem services accounts
	Monetary ecosystem services accounts
Ecosystem asset	Ecosystem asset accounts
Ocean	Ocean accounts
Species	Species accounts
Carbon	Carbon accounts
Urban	Urban accounts
Protected areas	Protected areas accounts
Forest	Forest accounts

multiple account types. In the AFE region, 3 countries are at Stage I, 4 at Stage II, and 4 at Stage III, indicating that while uptake is modest, a meaningful cluster of countries has progressed beyond initial experimentation. Three additional countries not listed in the Global Assessment have efforts towards compiling accounts, not necessarily in the SEEA framework. For that reason, we have added a new category, "Compiling (non-SEEA)", to the stage labels to reflect countries that have efforts towards compiling accounts but are not yet implementing the SEEA. Figure 1 visualizes the implementation status of the AFE countries.

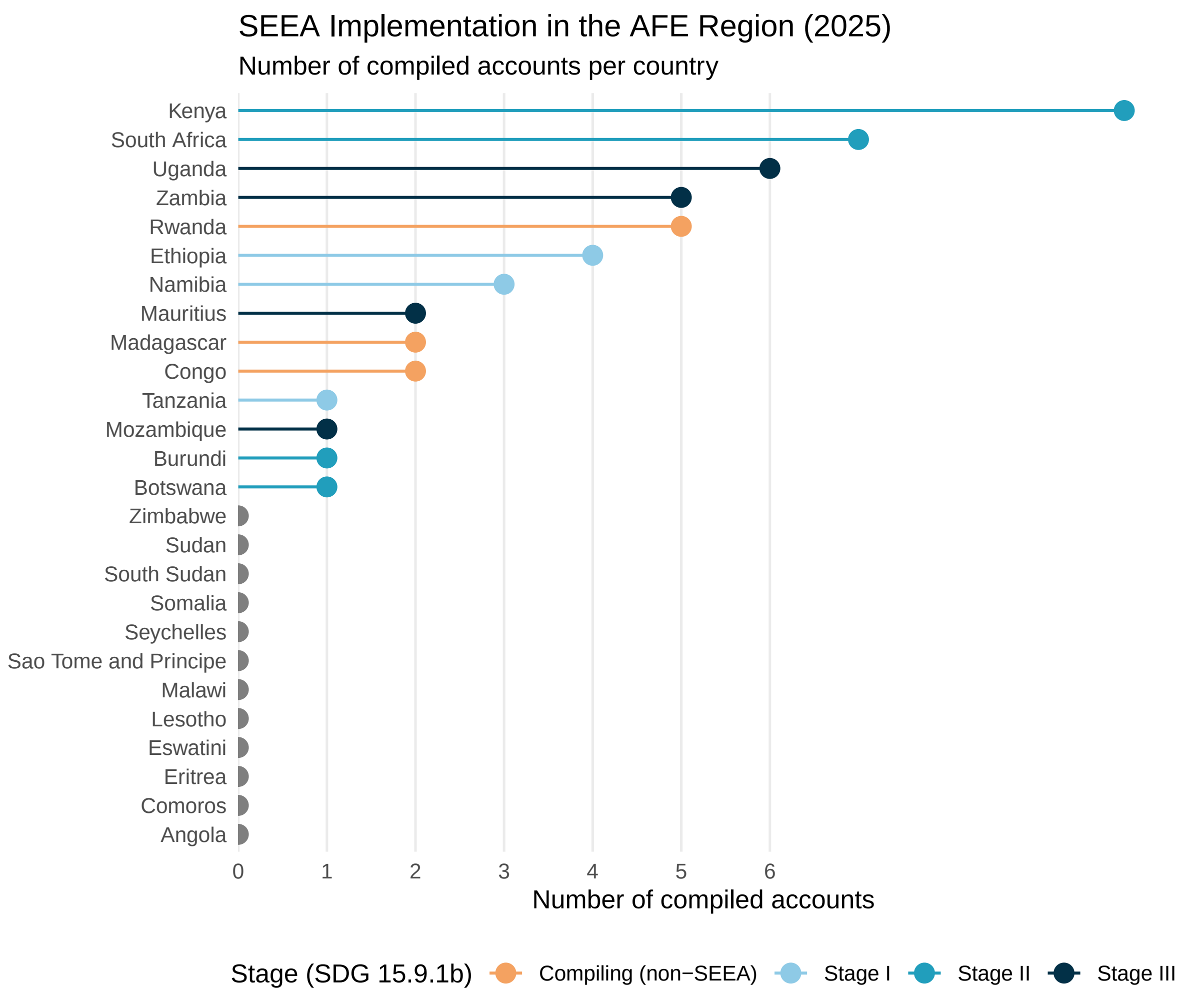

Figure 1: AFE countries by SEEA implementation stage and number of compiled accounts. Countries are sorted by compiled account count; non-implementing countries are shown in grey.

2.2 Account Coverage

Across all 11 implementing countries, a total of 41 account-compilations have been recorded covering 19 distinct account types out of the 36 tracked. The most widely compiled accounts are the Physical Supply and Use Tables (PSUTs) for water, compiled by 6 countries, followed by PSUTs for Energy and Ecosystem Extent Accounts, each compiled by 5 countries, reflecting their relative accessibility as entry-level accounts and their relevance to the region's resource management priorities. Physical asset accounts for land follow, compiled by 4 countries. Beyond this initial tier, uptake drops sharply: monetary accounts, which require additional institutional capacity and data integration, and ecosystem accounts appear in only 1 to 3 countries each, with thematic accounts for forests and protected areas at the same level.

Figure 2: Account compilation status across the 11 implementing AFE countries, grouped by account category. Countries are ordered left to right by total number of compiled accounts; accounts within each group are ordered by frequency of compilation.

Table 2 provides an aggregate view of account group coverage, showing the number of distinct

Table 2: Summary of SEEA account coverage by account group across the 11 implementing AFE countries.

Account Group	Accounts tracked	Compilations recorded	Countries co
SEEA Central Framework accounts	25	29	
SEEA Ecosystem accounts	5	19	
Thematic accounts	6	2	

accounts within each group, how many individual account-compilations have been recorded across implementing countries, and how many countries have compiled at least one account in each group.

3 General findings and policy insights

3.1 Water accounts

Water accounts are the most widely compiled component of SEEA implementation in the AFE region and the most directly actionable for policy. Six countries have compiled physical supply and use tables for water, and Rwanda has gone further to produce monetary supply and use tables, the only such account in Africa, linking water flows to GDP through sector-level water productivity estimates. Rwanda's monetary accounts reveal that agriculture accounts for over 96 percent of total freshwater abstraction but achieves a water productivity of only 118 Rwf per cubic metre (118 Rwf/m^3), compared to 6,236 Rwf/m^3 for mining and 9,300-11,600 Rwf/m^3 for services; GDP grew 25 percent while water use grew only 10 percent between 2012 and 2015, a decoupling signal that would be invisible in standard national accounts (Government of Rwanda, 2019b). This means that the economy requires progressively less water to produce the same amount of GDP, which suggests that the economy is becoming more efficient in using water. Uganda's tenth-iteration water accounts further extend this approach. National water productivity reached UGX 5,557 per m^3 consumed in 2024 (84 percent above the 2015 baseline) but oil sector abstraction grew 113.9 percent year-on-year, signalling an emerging demand that the time-series accounts are uniquely positioned to track (Uganda Bureau of Statistics, 2025).

A consistent finding across physical water accounts is that national-level aggregates conceal stark spatial heterogeneity. Kenya's accounts document a tenfold range in per-capita water availability across basins, from 8 m^3 per person per year in Lake Victoria South to 91 m^3 in the Tana basin, while the Rift Valley shows groundwater abstraction exceeding 100 percent of sustainable yield (World Bank & Prime Africa Consult, 2025). South Africa's Strategic Water Source Area (SWSA) accounts demonstrate that three of the 22 SWSAs, which together supply approximately two thirds of national economic activity from just 8.2 percent of land area, have already fallen below the 60 percent natural land cover threshold necessary for functional water production (Statistics South Africa, 2023). Zambia frames water not only as a production input but as a macroeconomic vulnerability, because Lake Kariba fell below 15 percent of capacity in 2016 and again in 2024, in each case triggering nationwide load-shedding, while the 2023/24 El Niño drought simultaneously crippled hydropower generation and destroyed the maize harvest, illustrating the dual pathway through which hydrological shocks propagate into the economy (Chisanga et al., 2026).

Distribution losses and infrastructure condition are a recurring theme across reviewed accounts. Mauritius, which maintains the most institutionalized water accounting programme in the region (fifth consecutive edition), records a rate of 64.6 percent of unaccounted-for water in the Central Water Authority network, which was up from 56.4 percent in 2013, meaning nearly two-thirds of treated potable water is lost before reaching end users, despite 99.8 percent national access to safely managed drinking water (Statistics Mauritius, 2024). Rwanda's Water and Sanitation Corporation (WASAC) utility loses 41 percent of output in distribution losses (US$8.7 million per year in foregone revenue) and Uganda's supply chain records distribution losses of 34.8 percent. Taken together, these figures, directly readable from water supply and use tables, provide the economic argument for infrastructure investment.

3.2 Land accounts

Land accounts are the most universal entry point into SEEA implementation, because every country in this review has compiled at least one form of land cover or land use change account. The consistent finding throughout the implementations suggests accelerating cropland expansion at the expense of natural vegetation, but the policy implications depend critically on whether accounts track conversion pathways rather than just net change. In Ethiopia, cropland expanded from 21.7 to 25.9 million hectares between 2013 and 2022, which is a 20 percent increase, while built-up area grew 50 percent, yet the accounts caution that an observed 22 percent increase in forest cover over the same period reflects definitional and source-data differences as much as any ecological recovery (Federal Democratic Republic of Ethiopia, 2026). Rwanda presents the sharpest land transition in the region, where cropland rose from 24.2 to 51.7 percent of national territory between 1990 and 2015 as sparse forest lost 673,137 hectares and average farm parcels shrank to 0.184 hectares. This land fragmentation threshold limits mechanization and economic transformation (Government of Rwanda, 2018). The intensity of land use change in Rwanda is reflected in formal land market data, where transaction prices averaged 35 million Rwf per hectare in urban areas.

South Africa's accounts occupy the most technically advanced position in the region, combining ecosystem extent indices, classification hierarchies, and turnover metrics over three decades. A central finding is that while natural land nationally remained broadly stable (+0.8 percent), pivot irrigation expanded 221 percent between 1990 and 2018, the mine footprint grew 15.9 percent, and 18 percent of ecosystem types have already crossed the 60 percent natural cover threshold associated with ecosystem dysfunction (Statistics South Africa, 2020). Kenya's accounts contribute a methodologically important approach to transition analysis. Rather than presenting static totals, the land accounts compute a spatial backbone for county-level disaggregation and map conversion flows. These show a net forest loss of 342,793 hectares across 1990-2024, making it possible to identify where interventions such as restoration, conservation employment, or zoning changes would be most effective (Turpie, Wilson, Hickinbotham, et al., 2025). Zambia's accounts add a sub-national degradation lens, with the National Ecosystem Degradation Index (NEDI) showing that Eastern Province (5.25 percent) is degrading at nearly four times the national average (1.44 percent), driven by a woodland-to-cropland conversion pattern distinct from more stable provinces in the north-west (Chisanga et al., 2026). Across the region, the key advancement needed is the addition of monetary land accounts: currently, no country has compiled formal monetary values for land assets within the SEEA framework, limiting the ability to link land conversion to GDP or wealth accounts.

3.3 Energy accounts

Energy accounts remain the least compiled component in the AFE review, but where they exist they reveal a structural tension between physical and monetary energy realities that is directly relevant to the energy transition. Kenya's monetary energy supply and use tables, which are among the first in Sub-Saharan Africa, show that biomass dominates total primary energy supply in physical terms (1.2 million TJ) but that household monetary expenditures are concentrated on electricity and petroleum products (total supply value KES 580.5 billion; household spending KES 749.8 billion), with biomass essentially unpriced in market terms (World Bank & Prime Africa Consult, 2025). This divergence between physical and monetary accounts is policy-critical, because it means that energy subsidies, fuel taxes, and electricity tariffs are calibrated to a monetized minority of total energy use, while the majority (rural biomass) remains outside the price and fiscal system.

South Africa's accounts represent the region's most developed energy time series, with coal supplying 72.2 percent of total domestic energy but oil being 93.1 percent import-dependent, creating structural resilience asymmetries (Statistics South Africa, 2025). Mining and quarrying consumes 1.21 TJ per million Rand of gross value added, taking the spot as the most energy-intensive sector, and household biofuels account for 46.5 percent of residential energy, a share that has barely changed over the decade of accounts available. The biomass-forest nexus is especially pronounced for the DRC (94 percent of national energy from biomass), Uganda (78 percent), and Zambia (78 percent), where fuelwood harvesting is a direct driver of deforestation quantified in the forest and ecosystem accounts. DRC's forest accounts document a 709 percent increase in fuelwood harvesting in forest-farm mosaic areas over 2000-2020, making the energy-land connection explicit in physical terms and providing a basis for costing energy transition investments (Turpie, Wilson, Brühl, et al., 2025).

3.4 Ecosystem accounts

Ecosystem accounts are the most analytically powerful component of SEEA but also the most demanding to compile, and the AFE region shows a clear gradient from comprehensive forest accounts in the DRC to largely physical land-ecosystem hybrids in Ethiopia and Madagascar. Where monetary ecosystem service valuations have been produced, they reveal contributions to economic welfare that national accounts systematically miss. Kenya's monetary ecosystem service accounts estimate total services at KSh 247 billion per year, which is equivalent to 1.8 percent of GDP, with livestock grazing being the dominant service (KSh 185 billion), followed by nature-based tourism (KSh 38 billion) and sediment retention (KSh 24 billion) (Turpie, Wilson, Hickinbotham, et al., 2025). South Africa's biodiversity economy accounts estimate that biodiversity tourism contributed R27.7 billion to GDP (0.4-0.5 percent) and supported 91,836 direct jobs in 2019, figures that can be tracked over time to assess the economic case for conservation and mixed agroforestry systems (Statistics South Africa, 2024).

The DRC's accounts illustrate both the ceiling of what is currently achievable and the challenge of translating global ecological value into national incentive. Total forest ecosystem services were valued at US$706 billion per year in 2020, including global climate regulation at the social cost of carbon (US$79.86/tonne CO), but the domestically captured portion, excluding carbon externalities priced globally, was only US$3.3 billion per year (Turpie, Wilson, Brühl, et al., 2025). The gap between global and domestic value is the economic explanation for continued deforestation. The deforestation rate doubled from 0.5 to 0.8 percent per year between 2000 and 2020, and forest depletion costs grew 113 percent between 2010 and 2020, yet these losses accrue to livelihoods too diffuse and informal to be captured in national accounts. The accounts also document that bushmeat extraction is 64 percent above sustainable yield, which is a physical condition indicator with direct food security implications, and that the total forest asset value declined from US$12.0 trillion in 2000 to US$10.3 trillion in 2020; a national wealth loss of US$1.7 trillion that would appear in a SEEA-compliant balance sheet.

Physical ecosystem accounts in Ethiopia and Rwanda demonstrate how regulating services, which are often omitted from monetary analyses, can be quantified at hydrological scales relevant to investment decisions. Ethiopia's accounts estimate that 29.7 billion tonnes of soil erosion was avoided annually by existing land cover, but that baseflow declined by 1.3 percent and flood mitigation capacity by 2.7 percent over the accounting period, and that Addis Ababa's rapid urbanization reduced catchment water yield by 2,691 m^3 per hectare (Rutebuka et al., 2025). Rwanda's ecosystem accounts track soil erosion increasing 54 percent, sediment export, 123 percent, and quickflow, 35 percent, between 1990 and 2015. These are quantified degradation trends driven by the same cropland expansion documented in the land accounts, while baseflow declined 11 percent nationally and 29-32 percent in peri-urban districts (Government of Rwanda, 2019a). These accounts close the feedback loop between land use change and hydrological service provision in a way that land accounts alone cannot.

3.5 Common Data Gaps and Systemic Recommendations

A detailed review of the reports underpinning the Annex reveals that data gaps and institutional constraints follow recurring patterns across countries, but with important country-specific dimensions that inform where targeted support would yield the greatest returns. The following subsections organize these findings around six themes.

3.5.1 Data Infrastructure and Monitoring Gaps

Measurement infrastructure for the primary inputs to SEEA accounts remains underdeveloped across the region. In Zambia, the water accounts explicitly acknowledge that a fully SEEA-Water-compliant national account has not yet been institutionalized, with groundwater abstraction "largely unmetered" and monitoring networks "remaining sparse" (Chisanga et al., 2026). Kenya identifies the Rift Valley basin as a groundwater stress hotspot where the use-to-availability ratio exceeds 100 percent, yet no remediation monitoring is in place. The accounts also state that water quality accounts, covering nutrient pollution, dissolved oxygen, and fluoride, are urgently needed but not yet compiled (World Bank & Prime Africa Consult, 2025). Rwanda projects demand growing tenfold by 2040 driven by irrigation, yet the current monitoring network was designed for far lower abstraction rates, creating a foreseeable data gap at the moment when it would be most policy-critical (Government of Rwanda, 2019b). Distribution losses, which are a key diagnostic for infrastructure investment, are substantial but inconsistently tracked. For example, Mauritius records 64.6 percent unaccounted-for water in the CWA network, worsening from 56.4 percent in 2013, yet no monetary cost-of-leakage calculation has been compiled (Statistics Mauritius, 2024); the Water And Sanitation Corporation (WASAC) in Rwanda achieves only 75 percent cost recovery, with non-revenue water at 41 percent (US$8.7 million per year) (Government of Rwanda, 2019b); and Uganda's supply chain records distribution losses of 34.8 percent in a system where daily consumption per-household is already well below basic-needs thresholds (Uganda Bureau of Statistics, 2025). In the DRC, poverty and inequality data remain outdated, with the most recent Gini coefficient (42.1) dating from 2012 and poverty headcount from 2004, while gross national savings data for 2000 are simply absent, preventing the calculation of adjusted net savings for that year (Turpie, Wilson, Brühl, et al., 2025). Energy monitoring in DRC, Uganda, and Zambia all face the same structural gap, related to informal and subsistence biomass flows, which represent 78 percent of national energy in Uganda and Zambia and 94 percent in the DRC. In all three cases, these are estimated rather than measured, introducing material uncertainty into energy and forest supply-use balances (Chisanga et al., 2026; Turpie, Wilson, Brühl, et al., 2025; Uganda Bureau of Statistics, 2020).

3.5.2 Classification and Cross-Country Comparability

The forest cover data across the region illustrate the depth of the classification problem. Ethiopia's accounts document a 22 percent increase in total forest cover between 2013 and 2022. However, the report explicitly cautions that "differences in data sources, forest definitions, and canopy cover sensitivity to rainfall make it difficult to apportion change between ecological dynamics and reclassification," putting into question whether observed gains represent true restoration or methodological nuances (Federal Democratic Republic of Ethiopia, 2026). Zambia's ecosystem extent accounts use MODIS satellite data with classification boundaries that differ from national land cover data, so that some biome extent totals exceed what would be expected from the national territory, and the accounts interpret percentage changes as more reliable than absolute areas (Chisanga et al., 2026). The DRC uses fifteen ecosystem types including forest-farm mosaic as a distinct class; this level of granularity enables important policy insights. For example, a mosaic expansion of 83 percent in 2000-2010 and 96 percent in 2010-2020 reveal an important agricultural frontier dynamic. However, at the same time it makes comparison with neighbouring countries' simpler classification schemes difficult (Turpie, Wilson, Brühl, et al., 2025). Madagascar, Ethiopia, and Zambia all rely on remotely sensed vegetation indices as primary condition indicators, without consistent field validation or cross-country calibration of thresholds. Kenya's ecosystem condition accounts find that no forest ecosystem achieves a condition score above 60 percent nationally. This is a finding that may reflect genuine degradation but also the choice of reference state and indicator sensitivity (Turpie, Wilson, Hickinbotham, et al., 2025). Harmonizing forest and land cover classification systems across the AFE region, even through common reporting categories alongside national definitions, would substantially improve the analytical value of cross-country comparison.

3.5.3 Monetary Valuation Gaps

The absence of monetary ecosystem service accounts remains the most consequential gap for the policy applications that national capital accounting is intended to support. Ethiopia's accounts are fully physical across all seven ecosystem services, with monetary valuation explicitly deferred to a future phase (Rutebuka et al., 2025). Madagascar likewise has no monetary accounts. The economic analysis of hydropower losses, agricultural productivity costs, and carbon opportunity costs was computed as part of a companion report with no official publication (Vogl et al., 2022). Rwanda's ecosystem accounts cover soil erosion, sediment export, water yield, and quickflow in physical units only, despite Rwanda being one of two countries in the region with monetary water supply and use tables (Government of Rwanda, 2019a). In Zambia, the only monetary ecosystem services estimate available is the net present value of carbon stock loss, which amounts to USD 46 million over 2018-2023, while the Bangweulu wetlands ecosystem services (US$11.7 million per year) and the fish production sector (US$422.6 million in 2023) remain outside the formal NCA monetary framework (Chisanga et al., 2026). The DRC's forest accounts, while the most comprehensive in the region for forest ecosystems, illustrate the valuation challenge. The total monetary value of forest ecosystem services including global climate regulation was US$706 billion per year in 2020, which is a figure that includes the social cost of carbon at US$79.86 per tonne CO , but the domestically captured benefit (excluding carbon externalities) was only US$3.3 billion per year, creating a vast gap between global value and national incentives (Turpie, Wilson, Brühl, et al., 2025). Across the region, the most consistently missing services in monetary terms are pollination, which is absent everywhere, freshwater provisioning to households, which remains unpriced, flood damage avoided, which is proposed but not compiled, and the contribution of forests to agricultural production through rainfall maintenance and soil protection.

3.5.4 Spatial Disaggregation Needs

National-level aggregates systematically mask the localized pressures that are most relevant to investment targeting. Kenya's water accounts reveal a tenfold range in per-capita water availability across basins. From 8 m^3 per person per year in Lake Victoria South to 91 m^3 per person per year in Tana basin, which is a difference that is invisible in any national-level figure (World Bank & Prime Africa Consult, 2025). Zambia's provincial disaggregation shows that Eastern Province's ecosystem degradation index (NEDI = 5.25 percent) is nearly four times the national average (1.44 percent), with distinct conversion patterns, such as woodland-to-cropland, that differ fundamentally from those in the more stable North-Western Province (NEDI = 0.12 percent) (Chisanga et al., 2026). South Africa's Strategic Water Source Areas (SWSA) accounts demonstrate that three specific catchments have already fallen below the 60 percent natural land threshold considered necessary for functional water production. This is a finding with direct implications for Gauteng's water security, which would be invisible in a national land cover account showing stable natural vegetation (Statistics South Africa, 2023). Rwanda's ecosystem accounts document that baseflow declined in every district between 1990 and 2015, with the largest losses in peri-urban Kigali (Kicukiro with 32 percent and Ngoma with 29 percent), yet the national aggregate shows only an 11 percent decline (Government of Rwanda, 2019a). For the DRC, provincial disaggregation reveals that Kwilu province lost 720,000 hectares of lowland forest, representing the largest absolute loss in the country, while Kinshasa province lost 67 percent of its forest cover, suggesting that proximity to population centers drives deforestation patterns that differ from remote interior dynamics (Turpie, Wilson, Brühl, et al., 2025). The consistent message is that national accounts need to be complemented by sub-national or catchment-level disaggregation to serve as decision tools for sector investment, conservation targeting, or early warning systems.

3.5.5 Institutionalization and Continuity

The most persistent structural challenge is the transition from one-off project-funded accounts to permanent, institutionalized statistical products with public budget provisioning. The DRC's forest accounts state directly that "the DRC needs to ensure the institutionalization of data-systems" and that "the future compilation of natural capital accounts should ideally be led by the national statistics agency, working in partnership with the relevant financial, environmental, and sectoral ministries" (Turpie, Wilson, Brühl, et al., 2025). Zambia's water accounts acknowledge that institutionalization within the formal statistical system remains incomplete. Uganda's forest accounts project that the sustainable wood supply will be fully depleted by approximately 2025 under business-as-usual conditions, yet there is no standing mechanism within the national statistical system to trigger a policy response based on findings of the accounts. Currently, the accounts represent a research output rather than an operational monitoring tool (Uganda Bureau of Statistics, 2020). Against this background, Mauritius stands as the most instructive positive example in the region, because its water accounts are in their fifth edition, updated regularly, produced by Statistics Mauritius in collaboration with three operational agencies (the Water Resources Unit, the Central Water Authority, and the Waste Management Authority), and are grounded in data systems that are maintained independently of any single project cycle (Statistics Mauritius, 2024). Uganda's water accounts, now in their tenth iteration spanning 2015-2024, provide a second model of systematic series maintenance. Ethiopia's NCA Roadmap, developed alongside the first accounts, explicitly anticipates a transition to institutionalized production and frames the current work as a foundation rather than an endpoint (Federal Democratic Republic of Ethiopia, 2026). These examples suggest that institutionalization depends less on technical capacity than on the establishment of formal inter-agency data-sharing agreements, clear mandates within the national statistical plan, and budget lines that are independent of donor project cycles.

3.5.6 Thematic and Structural Gaps

Several categories of natural capital that are highly relevant to the AFE region are entirely absent from current account coverage. Marine and ocean accounts are compiled by no country in the region, despite the economic importance of Indian Ocean fisheries for Kenya, Tanzania, Madagascar, and Mozambique, and the formal significance of South Africa's 200 km^2 EEZ. The only coastal ecosystem accounts in the region are South Africa's experimental estuary accounts, which reveal that only 4 percent of assessed estuaries have good fish condition, sawfish are now extinct in South Africa, and dusky kob spawner biomass is below 2 percent of pristine levels (Niekerk et al., 2020). These are findings that suggest the absence of formal coastal accounts elsewhere is concealing analogous, if unquantified, degradation trends. Biodiversity accounts are present only in experimental form in South Africa (biodiversity tourism, 2013-2019) and the DRC (IBIS-model biodiversity status index, declining from 68.8 to 61.4 percent between 2000 and 2020). Kenya documents that wildlife populations have fallen to 27.3 percent of their 1977-1980 levels and are projected below 10 percent by 2050 under business-as-usual (Turpie, Wilson, Hickinbotham, et al., 2025). Mineral accounts are absent across the entire AFE region, despite South Africa's mine footprint growing 15.9 percent (1990-2014), Zambia's Copperbelt generating documented water quality hotspots from acid mine drainage, and the DRC deriving substantial government revenue from mining activities while forest depletion accelerates. Fisheries and aquaculture accounts are similarly absent, while Zambia's fish production sector is valued at US$422.6 million (2023) with aquaculture growing 112 percent over five years. Nevertheless, no formal ecosystem account frames this activity within a natural capital balance (Chisanga et al., 2026).

3.5.7 Priority Areas for Capacity Development

The review of country-level accounts reveals a clear two-tier structure in AFE's natural capital accounting. A leading group, consisting of Kenya, South Africa, Uganda, and Zambia, has built programmes spanning multiple account types and framework components, and represents the natural candidates for peer learning, south-south knowledge exchange, and cross-border data harmonization on shared basins and ecosystems. A second tier, consisting of Ethiopia, Madagascar, and Mauritius, has made substantive starts but remains largely anchored in physical Central Framework accounts, suggesting that targeted support to expand into monetary valuation and ecosystem accounting would yield the highest marginal returns for existing institutional investments.

The most important gaps, measured by their potential to change policy decisions, lie in monetary accounts. The capacity and statistical infrastructure requirements for full monetary ecosystem service valuation are substantially higher than for physical accounts, but the policy arguments they enable; for example,cost-benefit comparisons with infrastructure alternatives, pricing of water and forest services, fiscal case for conservation investment; are more powerful in turn. Within ecosystem accounts, condition accounts and monetary ecosystem services accounts are the priority extensions for countries already compiling extent accounts. Thematically, ocean and coastal accounts, biodiversity accounts, and mineral accounts are essentially absent from the region and represent a new frontier of potential account development, particularly relevant to the region's coastal states and mineral-resource-dependent economies.

4 Annex: Country-Level Policy Insights from Available Accounts

This annex draws directly on published reports and supporting data files for each country to extract specific indicators, facts, and data insights relevant to planning, policy, and World Bank operations. Entries are organized by country and, within each country, by the SEEA accounts that have been compiled, following the account order in Table 1. Where monetary valuation of ecosystem services has not yet been completed, this is noted explicitly together with the magnitude of the physical flows involved and available economic proxy data, to indicate the scale of what remains to be valued. The annex is intended as a reference for analysis.

4.1 Democratic Republic of Congo (DRC)

The Democratic Republic of Congo has compiled five SEEA Ecosystem Accounting accounts, consisting of ecosystem extent, ecosystem condition, physical ecosystem services, monetary ecosystem services, and ecosystem asset accounts, which are all presented in a single comprehensive report published by the World Bank Group in August 2025 and covering the period 2000 to 2020 (Turpie, Wilson, Brühl, et al., 2025). The accounts are restricted to forest ecosystems, reflecting the DRC's status as the custodian of the world's second-largest tropical forest biome. They were produced by Anchor Environmental Consultants under the leadership of Dr Jane Turpie, with World Bank project leadership from Kanta Kumari Rigaud, Tsegaye Ginbo Gatiso, and colleagues, and with financial support from CAFI and the Climate Investment Funds. The macroeconomic context is stark. Results show that the forestry sector contributes approximately 9 percent of GDP (down from 11 percent in 1990). Moreover, 94 percent of the country's total energy supply comes from forest biomass (98.8 percent of household energy), and approximately 90 percent of logging activity in 2014 was informal, small-scale, or illegal. With a population of approximately 96 million in 2020 and a poverty headcount of 63.9 percent (2012 data), the DRC's forests simultaneously constitute its largest natural capital asset and the primary energy source for tens of millions of households.

4.1.1 Ecosystem Extent Accounts (SEEA Ecosystem Accounting)

The ecosystem extent accounts use a 15-class typology to document forest and land cover change from 2000 to 2020 across the DRC's 230 million hectares (Turpie, Wilson, Brühl, et al., 2025). Lowland forest, which is the dominant class, covered 104,336,681 ha (45.2 percent) in 2000 and declined to 97,467,643 ha (42.3 percent) by 2020, a net loss of 6,869,000 ha (6.6 percent) over two decades. Submontane forest fell from 3,096,792 to 2,884,494 ha (−6.9 percent) and montane forest from 1,900,909 to 1,612,987 ha (−15.1 percent). Total forest cover lost across all types between 2000 and 2020 was 11,568,000 ha. The rate of loss accelerated sharply, reflecting an annual deforestation rate of approximately 0.5 percent per year in the first decade (2000-2010), and rising to above 0.8 percent per year in the second decade (2010-2020), more than doubling. The most revealing transition is the expansion of forest-farm mosaic-degraded, mixed-use agricultural-forest landscapes, that changed from 2,777,041 ha (1.2 percent) in 2000 to 9,970,589 ha (4.3 percent) in 2020 (+259 percent over 20 years), with growth of 83 percent in the first decade and 96 percent in the second. Cropland expanded from 1,451,609 ha (0.6 percent) to 3,592,903 ha (1.6 percent), a 147 percent increase. Human-transformed ecosystems collectively grew from 2.2 to 6.8 percent of national land cover. At the provincial level, Kwilu suffered the largest absolute lowland forest loss (720,000 ha, 2000-2020), while Kinshasa province lost 67 percent of its forest cover and Kasai Oriental 50 percent, documenting the acute forest pressure around urban centres and agricultural frontier zones.

4.1.2 Ecosystem Condition Accounts (SEEA Ecosystem Accounting)

The condition accounts track biodiversity status using the IBIS (Integrated Biodiversity Assessment System) model, which estimates the percentage of original species populations remaining given observed habitat change. National biodiversity status declined from 68.8

percent in 2000 to 65.5 percent in 2010 and 61.4 percent in 2020, indicating a continued erosion of ecological integrity even in a country where over 60 percent of territory remains forested. Lowland forest, which is the biologically richest category, fell from 69.2 to 61.4 percent. Carbon condition tells a different story, where net carbon retention increased by 2.6 percent (2000-2010) and 2.4 percent (2010-2020) due to a CO fertilization effect on biomass accumulation in intact forests, even as total forest area declined. The accounts call attention to an acute overexploitation in the bushmeat sub-sector. Extraction rates were 60 percent above sustainable yield in 2000 and had risen to 64 percent above sustainable yield by 2020 (Turpie, Wilson, Brühl, et al., 2025). Bushmeat income represents approximately 32 percent of annual household income in some forest-dependent communities such as the Rubi-Tele Hunting Domain, making its overexploitation both an ecological risk and a livelihood vulnerability.

4.1.3 Physical and Monetary Ecosystem Services Accounts (SEEA Ecosystem Accounting)

Carbon stocks. The accounts estimate total carbon stored in all DRC forest ecosystem types in 2020 at 24,230 million tonnes of biomass carbon plus 31,410 million tonnes of soil organic carbon, for a total of 55,640 million tonnes of carbon-equivalent to approximately 204,000 million tonnes of CO , which is described in the accounts as "almost six times greater than the amount of energy-related CO emissions globally in 2020" (Table O.2, Turpie, Wilson, Brühl, et al. (2025)). Peat forests alone store 19,576.9 million tonnes of soil organic carbon at an average density of 1,917 tonnes per hectare, which is the highest density of any ecosystem type globally. The accounts note that DRC's forests sequester carbon at a mean net rate of 1.13 tonnes of carbon per hectare per year in non-mangrove forests. Moreover, at the social cost of carbon applied in the accounts (USD 79.86/tCO e in 2020), the global climate regulation service of DRC forests was valued at CDF 1,983,553 billion (US$703,387 million) per year in 2020, which is by far the largest single ecosystem service value.

Monetary ecosystem services. Total annual monetary supply of forest ecosystem services reached CDF 1,991,451 billion (US$706,188 million) in 2020, driven almost entirely by the global climate regulation value (Table O.3, Turpie, Wilson, Brühl, et al. (2025)). Excluding the global carbon externality, the domestically captured value was CDF 9,434 billion (US$3,345 million) per year in 2020, consisting of timber US$1,597 million; other wild resources (bushmeat, wild plants, other) US$476 million; sediment retention services US$709 million; and nature-based tourism US$20 million. Sediment export, an ecosystem disservice, increased approximately 71 percent between 2000 and 2020, reflecting watershed degradation from forest conversion. Fuelwood harvesting from forest-farm mosaics increased 709 percent between 2000 and 2020 as the agricultural frontier expanded, reflecting both population pressure and the near-total dependence of households on biomass for energy.

4.1.4 Ecosystem Asset Accounts (SEEA Ecosystem Accounting)

The ecosystem asset accounts value DRC's forests at discounted present values of future service flows, calibrated against reference international prices. Total asset value increased from CDF 15,359 trillion (US$5,446 billion) in 2000 to CDF 29,143 trillion (US$10,335 billion) in 2020, almost doubling in nominal international dollar terms, largely due to the rising global valuation of carbon (Tables 9.1-9.2, Turpie, Wilson, Brühl, et al. (2025)). Excluding the global carbon premium, the asset value was US$21.6 billion in 2000 rising to US$42.0 billion in 2020. Despite this nominal growth, per capita asset value declined by 4 percent overall between 2000 and 2020, as population growth (from 48.6 million to 96 million) outpaced the expansion in total forest value. Montane forests experienced the sharpest per capita decline (−18 percent), reflecting both forest loss and population concentration in the highland zones where montane forests are found.

Adjusted National Savings and sustainability. The accounts link forest depletion to national accounts through an Adjusted Net National Income (ANNI) framework (Table 10.1, Turpie, Wilson, Brühl, et al. (2025)). Forest depletion increased from CDF 818,896 million in 2000 to CDF 2,256,000 million in 2020, equivalent to a 113 percent increase in the 2010-2020 decade alone, eroding the apparent GDP and GNI growth. The accounts calculate that taking into account trajectories of unsustainable harvesting, the forest asset value in 2020 was CDF 19,583 billion (US$6.9 billion) less than it would have been if all harvests had been sustainable. Oil production is projected to decline permanently from 2026 onward, making the development of carbon finance, REDD+, and sustainable forest economy mechanisms an increasingly urgent fiscal priority for the DRC government. The accounts explicitly call for institutionalizing NCA compilation within the Institut National de la Statistique and for developing five-yearly update cycles to track trends in this rapidly changing landscape.

4.2 Ethiopia

Ethiopia has compiled four SEEA accounts, comprising physical and monetary asset accounts for land (SEEA Central Framework), and ecosystem extent and ecosystem condition accounts (SEEA Ecosystem Accounting). These were produced through a government-led Technical Working Group supported by the World Bank and the United Nations Statistics Division, bringing Ethiopia into the SEEA implementation programme. The accounts cover the period 2013 to 2022 and are disaggregated to the level of regions and major river basins, including the Abay (Blue Nile) and Awash basins. Monetary valuation of ecosystem services has not yet been undertaken. The ecosystem service accounts are presented in physical units only, and monetary ecosystem asset accounts remain a next step identified in Ethiopia's NCA Roadmap.

4.2.1 Land Asset Accounts (SEEA Central Framework)

The physical land asset accounts document Ethiopia's land cover across 13 classes for 2013 and 2022, recorded in hectares at national level and for the Abay and Awash river basin case studies (Tables 4.1-4.4 in Federal Democratic Republic of Ethiopia (2026)). The most consequential change was the expansion of cropland. Annual cropland grew by 20 percent, from 21.7 million hectares to 25.9 million hectares, while perennial cropland, dominated by coffee, khat, enset, fruit trees, and sugarcane, expanded by 50 percent, from approximately 727,000 to 1.09 million hectares. Together, croplands now cover roughly 27 million hectares, accounting for nearly a quarter of Ethiopia's total land area. This expansion came largely at the cost of shrublands, with open shrublands declining by 23 percent (approximately 7.7 million hectares) and closed shrublands by a similar margin. The built-up area grew by 50 percent, from 350,000 to 526,000 hectares, consistent with an industrial sector whose share of GDP rose from 12-15 percent in 2013 to 22-25 percent in 2022. Overall, only 52 percent of land cover remained in the same class across the two reference years, indicating a high rate of landscape change (Figure 4.1-4.2 in Federal Democratic Republic of Ethiopia (2026)).

Forest cover presents a more complex picture. Total forest area, combining dense, moderate, and sparse forest, increased from 22.3 million to 27.2 million hectares, a gain of 4.9 million hectares (22 percent). Dense forest grew by 13 percent (1.6 million ha) and moderate forest by 45 percent (3.6 million ha), while sparse forest declined by 13 percent (0.3 million ha). These gains are partly attributable to restoration and afforestation efforts including the Green Legacy Initiative, though the report cautions that differences in data sources, forest definitions, and canopy cover sensitivity to rainfall make it difficult to apportion change between ecological dynamics and reclassification (Section 4.5, Federal Democratic Republic of Ethiopia (2026)). Irrespective of cause, forest cover recorded in the accounts represents a critical natural asset. Ethiopia's forest sector is estimated to contribute approximately 13 percent of GDP when informal and subsistence uses are included (UNEP, 2016), or around 6 percent on a formal sector basis (MEFCC, 2017). In 2012-13, Ethiopian forests generated cash and in-kind economic benefits equivalent to USD 16.7 billion (approximately ETB 111 billion), underscoring the scale of forest dependency in the economy even before any formal monetization of ecosystem services.

The monetary context for land extends well beyond the forest sector. Between 2013 and 2022, Ethiopia's GDP grew from approximately USD 47.5 billion to USD 156 billion, which represents a near-tripling in nominal terms, while the agriculture, forestry, and fisheries aggregate expanded from USD 19.7 to USD 47.7 billion in nominal terms (or by 58 percent in real terms). Agriculture's GDP share fell from approximately 43 percent to 35 percent, reflecting structural transformation, yet the sector continued to employ roughly 70 percent of the workforce. Coffee alone generates USD 1.4 billion in annual export earnings, representing 35 percent of total export revenue. Working against these gains, land degradation is estimated to reduce Ethiopia's agricultural GDP by 2 to 6.75 percent annually (Nkonya et al., 2016), implying a foregone value in the range of USD 1-3 billion per year based on 2022 sector size.

This gap between the potential and actual productivity of Ethiopian land, visible spatially in the accounts, is precisely the type of information that monetary land asset accounts are designed to formalize and communicate to economic planners. Completing the transition from physical to monetary land accounts would give Ethiopia a tool to benchmark the cost of degradation against the returns from restoration investments such as those supported under the Ethiopia Strategic Investment Framework for Sustainable Land Management (ESIF-II).

4.2.2 Ecosystem Extent Accounts (SEEA Ecosystem Accounting)

The ecosystem extent accounts use land cover data as a proxy for ecosystem type distribution, following standard practice at this stage of SEEA-EA implementation. At the national level, the accounts confirm that shrublands, open and closed combined, remain the single largest ecosystem group, covering over 36 million hectares in 2022 despite a decline from 46 million in 2013. Croplands constitute the second largest category at 27 million hectares, followed by forests at 27.2 million hectares. Grasslands cover approximately 12.3 million hectares and wetlands and water bodies together account for a modest 1.5 million hectares. The spatial backbone of these accounts enables sub-national disaggregation that is policy-relevant in a country as ecologically diverse as Ethiopia. The Abay (Blue Nile) basin, for example, is 43 percent annual cropland, more than double the national share, making it the agricultural heartland of the country and the principal basin for soil erosion risk and watershed management investment (Figure 4.13, Federal Democratic Republic of Ethiopia (2026)). The Awash basin, by contrast, has 26 percent bare land, three times the national share, and the almost complete disappearance of closed grassland (from 91,000 ha in 2013 to just 66 ha in 2022) shows the severity of land degradation pressures in this basin. Both basins have Landscape Stability Index scores below 0.56, indicating that nearly half of the land changed class over nine years, a rate of change that warrants targeted monitoring.

The basin-level accounts also show the geographic concentration of Ethiopia's most productive ecosystems. Southwest Ethiopia, Benishangul-Gumuz, and Gambela consistently emerge as high-capacity zones across multiple ecosystem service indicators; such as soil retention, water regulation, and carbon storage; while Somali, Afar, and Addis Ababa rank persistently below national averages. This spatial heterogeneity means that national-level figures mask significant distributional realities, where investments directed at high-capacity ecosystems in the southwest may yield disproportionately higher returns per hectare than equivalent investments in the lowland east. Initiatives working on watershed management or nature-based solutions under ESIF-II or similar programs can use the basin-level disaggregation in the accounts to target priority catchments.

4.2.3 Ecosystem Condition and Services Accounts (SEEA Ecosystem Accounting)

Ethiopia's ecosystem condition accounts, compiled through experimental biophysical modelling, cover seven priority services, related to soil protection (avoided erosion), water quality regulation (avoided sediment), sustained dry-season water availability (baseflow), flood mitigation (avoided quickflow), water yield, carbon sequestration, and carbon stock (Rutebuka et al., 2025). All results are reported in physical units, while monetary valuation is explicitly deferred to a future phase. Here, we highlight the specific indicators available and their policy relevance, recognizing that converting these physical flows to monetary estimates would substantially strengthen the economic case for ecosystem protection and agroforestry mixed use practices.

Soil protection and agricultural productivity. Ethiopia's ecosystems avoided an estimated 29.7 billion tonnes of soil erosion in 2022, equivalent to 262 tonnes per hectare (Table 3, Rutebuka et al. (2025)). This service remained essentially stable nationally (+0.05 percent), but beneath this aggregate, declining per-hectare efficiency becomes a critical warning. Of the soil protection that directly benefits agriculture, recorded as a final service to the agricultural sector in the supply-use account, total use increased from 8.5 to 9.1 billion tonnes between 2013 and 2022, driven by cropland expansion, but the service delivered per hectare of cropland fell from 411 to 338 tonnes per hectare (Table 7, Rutebuka et al. (2025)). This 18 percent efficiency decline means that despite more land under cultivation, each hectare is receiving less protective cover from surrounding ecosystems, which is a pattern consistent with the conversion of high-functioning shrublands and forest edges into agricultural land. The Abbay basin alone accounts for nearly 4 billion tonnes of avoided erosion per year, over 40 percent of the national total (Table 8, Rutebuka et al. (2025)), making it the critical zone for soil conservation investment. The opportunity for monetary valuation is clear, because if even a fraction of the cost of land degradation (estimated at 2-6.75 percent of agricultural GDP annually) is attributable to erosion processes that ecosystem protection could prevent, the quantified soil protection service provides the physical basis for a cost-benefit analysis of watershed restoration and conservation investments.

Hydrological services and water security. National water yield reached 347 billion cubic metres in 2022, equivalent to 3,062 m^3 per hectare (+0.9 percent since 2013) (Table 3, Rutebuka et al. (2025)). However, the two hydrological regulation services that are most relevant to climate resilience both declined. Dry-season baseflow fell by 1.3 percent to 163 billion m^3 (1,440 m^3/ha), and flood mitigation capacity declined by 2.7 percent to 185 billion m^3 (1,635 m^3/ha). These are not offsetting changes, which indicate that while total water volume was maintained, the natural buffering capacity of Ethiopia's landscapes against both drought and flood is weakening. Addis Ababa is the most extreme case. Its flood mitigation value is negative (-2,691 m^3/ha) due to the dominance of impervious surfaces, meaning that the city now amplifies rather than attenuates peak flows compared to a baseline vegetation cover (Figure 10, Rutebuka et al. (2025)). The decline in flood regulation nationally is concentrated in basins where shrublands, which are the primary regulators of quickflow, have been converted to cropland. These physical indicators provide the building blocks for economic valuation. Avoided flood damage costs, irrigation water supply reliability, and hydropower generation stability (including the Grand Ethiopian Renaissance Dam) are all services that depend on maintaining adequate baseflow and reducing sedimentation. Southwest Ethiopia (5,039 m^3/ha) and Benishangul-Gumuz (3,527 m^3/ha) stand out as disproportionate contributors to dry-season water flows and represent high-priority zones for watershed investment from a water security perspective.

Carbon regulation and climate finance. Ethiopia's terrestrial carbon stock of approximately 34 billion tonnes of carbon remained virtually unchanged between 2013 and 2022 (average 300 t C/ha; Table 10, Rutebuka et al. (2025)). However, the annual carbon sequestration balance turned slightly negative at -0.006 tonnes C/ha/year, or approximately -0.65 million tonnes of carbon per year nationally (equivalent to approximately -2.4 million tonnes CO per year). This result is driven by land-use transitions. Shrubland conversion to cropland is generating net carbon emissions that are not fully offset by forest growth and cropland carbon uptake. Open shrubland alone emitted 194.8 million tonnes C/year, and closed shrubland a further 75.2 million tonnes C/year, against sequestration of 93.6 million tonnes C/year from moderate forests and 166.5 million tonnes C/year from annual croplands (Table 9, Rutebuka et al. (2025)). For climate finance purposes, this finding is significant, because Ethiopia's forests and croplands are demonstrably sequestering carbon at scale, but their contribution to net sequestration is being eroded by the continued conversion of shrublands. At indicative voluntary carbon market prices of USD 10-30 per tonne CO , the net positive sequestration services of forests and croplands (where they are growing) represent potential carbon revenue, while the net shrubland emission represents foregone revenue. Restoration of degraded shrublands, particularly in the Omo Gibe and Wabi Shebele basins, which registered the largest net emissions, is a direct investment opportunity that the carbon asset account can help prioritize. The accounts could support Ethiopia's participation in emerging markets under Article 6 of the Paris Agreement, as explicitly noted in the land accounts policy discussion.

Data gaps and priorities for monetary valuation. The ecosystem services accounts currently lack monetary estimates for any service. The following services have the most established valuation pathways and the strongest policy demand: soil protection valuation, via avoided cost of productivity loss from erosion, building on the 2-6.75 percent GDP damage estimate; water regulation services, via avoided flood damage and water supply reliability for hydropower and irrigation; carbon sequestration, via carbon market prices and NDC compliance values; and biomass provisioning (fuelwood and grazing), given that over 90 percent of Ethiopian households rely on wood as their primary energy source, implying a very large but unmeasured contribution to household welfare. Completion of monetary ecosystem service accounts would allow Ethiopia to formally report ecosystem contributions alongside GDP, to assess the cost-effectiveness of ESIF-II investments in ecosystem terms, and to access a wider range of climate and biodiversity finance instruments.

4.3 Kenya

Kenya has compiled the most comprehensive set of SEEA accounts in the AFE region, with ten accounts spanning five sectors and comprising water (physical supply and use tables and physical asset accounts), energy (physical and monetary supply and use tables), land (physical asset accounts), ecosystem accounts (extent, condition, physical services, and monetary services), and forest thematic accounts. These were produced through the Kenya National Bureau of Statistics (KNBS) in partnership with dedicated Technical Working Committees for each sector, under the National Plan for Advancing Environmental-Economic Accounting (NP-AEEA) 2025-2028, with financial and technical support from the World Bank Global Program on Sustainability. The accounts collectively provide a rare example of monetary ecosystem service valuation in the AFE region, spanning livestock production, sediment retention, and nature-based tourism. They position Kenya to integrate natural capital into fiscal planning through the Kenya Macroeconomic and Fiscal Model (KENMOD). As a background framing, 35 percent of Kenya's total wealth has been estimated to derive from natural capital including cropland and pastureland, and 18 percent of Kenya's GDP comes from counties where natural ecosystem services that underpin economic activity, sucha as water provision, soil fertility, pollination, climate regulation, are at high risk of degradation.

4.3.1 Water Supply and Use Tables and Asset Accounts (SEEA Central Framework)

Kenya's physical water accounts cover supply and use tables for 2021 and 2023, and a physical asset account at both national and basin levels, compiled as part of the NP-AEEA implementation (World Bank & Prime Africa Consult, 2025). In 2023, total national water supply was estimated at 5,716 million m^3, drawn predominantly from surface water, with groundwater abstraction increasing. Total precipitation over the accounting territory was close to 588,399 million m^3 in 2023, but most of this water was lost to evapotranspiration, leaving a fraction available for economic use. Electricity generation was the single largest sectoral user of water in the accounts, followed by households and agriculture. Non-revenue water losses, which is water produced but lost before reaching the consumer through physical leaks, unauthorized abstraction, or metering errors, declined modestly over the period, a positive trend given the cost implications for water utilities.

Basin-level disaggregation reveals highly uneven water availability and stress. In 2021, per capita water use ranged from just 8 m^3 per person per year in the Lake Victoria South basin to 91 m^3 per person per year in the Tana basin, against a national average of 26.6 m^3. This is a difference of more than tenfold within the same national territory (Table 3-5, World Bank & Prime Africa Consult (2025)). The Rift Valley basin showed the clearest signs of groundwater stress, with a groundwater use-to-availability ratio exceeding 100 percent, indicating that abstraction was surpassing the sustainable yield of the aquifer. This is a direct policy insight, showing that continued development in the Rift Valley without managed groundwater recharge or demand reduction risks compounding water insecurity in an already climate-stressed region. The Tana basin's high per-capita use reflects the dominance of hydropower and irrigation demands in this catchment, and the basin accounts provide the baseline for linking any future change in ecosystem condition (sediment loads, dry-season flow regulation) to the economic costs of infrastructure degradation. The accounts also take an initial step toward water quality accounts, documenting that nutrient pollution, low dissolved oxygen, and naturally high fluoride levels pose challenges across basins, and proposing a SEEA water quality account structure for future compilation (Section 3.4, World Bank & Prime Africa Consult (2025)).

4.3.2 Energy Supply and Use Tables (SEEA Central Framework)

Kenya has published physical energy supply and use tables annually in the KNBS Economic Survey since 2018. In 2025, the first monetary energy supply and use table (MSUT) was compiled for the year 2024, marking a significant milestone in the integration of physical and monetary information (Kenya National Bureau of Statistics, 2026a; World Bank & Prime Africa Consult, 2025). The physical accounts confirm Kenya's heavy reliance on renewable sources for grid electricity, which shows that geothermal (19,984 TJ) leads, followed by hydropower (13,071 TJ), wind (6,472 TJ), and solar (1,658 TJ). However, these grid flows are dwarfed by biomass in the overall energy balance, because firewood and charcoal together account for more than 1.2 million TJ of household energy use, underscoring that Kenya remains a biomass-dependent economy for cooking and heating despite its renewable electricity steps forward.

The monetary energy accounts reveal a fundamentally different picture from the physical accounts. Total monetary supply of energy products was valued at KES 580.5 billion in 2024, while recorded uses amounted to KES 672.9 billion. This constitutes a gap of KES 174 billion attributable to statistical residuals and the accumulation of distribution margins and taxes across the energy supply chain (Table 2-1, World Bank & Prime Africa Consult (2025)). Households dominate in monetary terms, with energy expenditures of approximately KES 749.8 billion. This stands in sharp contrast to the physical accounts, where households appear primarily as high-volume biomass users. In monetary terms, electricity and petroleum products drive household energy spending. Energy exports were valued at KES 55.2 billion. The monetary accounts expose that the predominance of biomass in physical terms masks a much larger electricity and petroleum market that has not historically been captured in environmental accounts. For initiatives advising on Kenya's energy transition, the accounts provide the first integrated picture of where value is generated and consumed along the energy chain.

4.3.3 Physical Asset Accounts for Land (SEEA Central Framework)

Kenya's physical land asset accounts, compiled by the World Bank GPS team with KNBS, use the System for Land-based Emissions Estimation in Kenya (SLEEK) programme's 30-metre satellite time series as the primary data source, supplemented by IMPRESS project land cover maps for 2015-2023 (Soulard, 2025). The accounts cover the period 1990-2024, providing one of the longer continuous land cover time series available for any country in the AFE region. Kenya's total land area is approximately 59 million hectares. Forest cover declined from 3,934,279 ha (6.65 percent of national area) in 1990 to 3,591,486 ha (6.07 percent) in 2024, a net loss of 342,793 ha over 34 years (Section 3.1, Kenya National Bureau of Statistics (2026b)). Using the FAO methodology restricted to land area only, the decline is from 6.78 percent to 6.19 percent over the same period. This reduction is concentrated in coastal and dryland forests, while montane forests within protected areas have been more stable. The ecosystem extent accounts extend this analysis to a broader ecosystem typology, showing that the 2002-2018 period saw annual cropland grow by approximately 1.4 million hectares (+29.5 percent), built-up areas double in extent, and all natural terrestrial ecosystem types decline in area.

4.3.4 Ecosystem Extent and Condition Accounts (SEEA Ecosystem Accounting)

Kenya's ecosystem extent accounts cover terrestrial and mangrove ecosystems for 2002 and 2018 (Turpie, Wilson, Hickinbotham, et al., 2025). Savanna ecosystem types, such as Acacia-Commiphora bushland, wooded grassland, and grassland, dominate the national landscape, collectively covering 66 percent of Kenya in 2018. Acacia-Commiphora bushland alone occupies approximately 46 percent of the national territory. Forests account for only 6 percent of land area, concentrated in the more humid, mountainous central and western regions. Between 2002 and 2018, all natural terrestrial ecosystem types declined in extent, with moist wooded grassland recording the steepest percentage loss (28.9 percent), followed by plantations (19.7 percent decline), Masai xeric shrubland (15.6 percent), montane forest (8.2 percent), and coastal wooded grassland (8.0 percent). These losses were driven principally by cropland expansion and, to a lesser extent, by built-up area growth, which are the two human-modified categories that together increased their national coverage from 8.8 to 11.3 percent over the accounting period (Table 3.2, Turpie, Wilson, Hickinbotham, et al. (2025)).

The ecosystem condition accounts assess the quality of forest and savanna ecosystems against pristine reference states, using remotely sensed indicators of tree height, tree cover, leaf area index, landscape connectivity, and vegetation indices. A key finding is that no forest ecosystem in Kenya achieved a condition score above 60 percent at the national level across either accounting year, indicating that a significant proportion of the forest estate is in degraded condition relative to a pristine reference state (Table 3.6, Turpie, Wilson, Hickinbotham, et al. (2025)). Within this picture, montane and western rainforests within protected areas remain comparatively intact, showing that approximately 30 percent of montane forest area was in good or pristine condition in 2018, while coastal forests fared worst, with no pristine areas remaining and over 80 percent falling into the moderate-to-extreme degradation classes. Over 40 percent of Kenya's remaining natural areas are estimated to be degraded (SPACES, 2023, as cited in Turpie, Wilson, Hickinbotham, et al. (2025)). The condition accounts provide the spatial diagnostic needed to prioritize conservation and restoration investment. The contrast between well-functioning montane water towers (Mt. Kenya, Aberdares, Mau Complex) and critically degraded coastal and dryland forests is visible in the accounts and can be directly linked to the distribution of ecosystem service values.

4.3.5 Monetary Ecosystem Services Accounts (SEEA Ecosystem Accounting)

Kenya's ecosystem service accounts are the most policy-relevant in the AFE region. They include both physical and monetary supply and use tables for three services compiled for 2002 and 2018; livestock production (provisioning), sediment retention (regulating), and ecosystem-based tourism (cultural) (Turpie, Wilson, Hickinbotham, et al., 2025). The monetary accounts value all three services in constant 2025 Kenyan Shillings, enabling direct comparison with GDP and SNA aggregates. In 2018, the total value of the three ecosystem services combined reached KSh 247 billion (approximately US$1,900 million), equivalent to 1.8 percent of GDP in that year (Tables 5.3-5.4, Turpie, Wilson, Hickinbotham, et al. (2025)). In 2002, the same services were valued at KSh 188 billion (approximately US$1,445 million), representing 2.8 percent of GDP. The decline in the ecosystem service share of GDP does not signal declining natural capital value. On the contrary, it reflects that Kenya's GDP grew substantially faster than the assessed ecosystem service flows over the period, a finding that itself points to incomplete accounts. Effectively, carbon, wood fuels, pollination, water regulation, and other services are not yet valued.

Livestock production. The ecosystem contribution to livestock production, which is quantified as the net value of meat and dairy production supported by natural grazing, valued at resource rent, increased by 32 percent from KSh 139,057 million (US$1,071 million) in 2002 to KSh 184,582 million (US$1,421 million) in 2018 (Table 5.4, Turpie, Wilson, Hickinbotham, et al. (2025)). Acacia-Commiphora bushland contributes 50-60 percent of this value, reflecting the dominance of extensive pastoralism in Kenya's arid and semi-arid lands. Livestock stocking supported by ecosystems increased from 5.6 million to 10.2 million Tropical Livestock Units (TLUs) over the period, with the highest densities concentrated in the Rift Valley and water tower regions. This makes the livestock service the largest single monetized ecosystem contribution to the Kenyan economy in the accounts, and one that is directly visible in the SNA through the agriculture sector's contribution to GDP.

Sediment retention. The monetary value of sediment retention services, estimated as the avoided cost of reservoir storage loss and instream water replacement, fell 19 percent from KSh 29,589 million (US$228 million) in 2002 to KSh 23,912 million (US$184 million) in 2018 (Table 4.6, Turpie, Wilson, Hickinbotham, et al. (2025)). Physically, total sediment retained by ecosystems declined from 883 million tonnes to 857 million tonnes over the same period (a 3 percent decline). Montane rainforests contributed the highest value in 2002 (32 percent of total), while Acacia-Commiphora bushland dominated in 2018 (35 percent). The accounts note that the apparent monetary decline likely understates the true service, as expanding demand for formal water supply within reservoir catchments was not fully captured in data available for the accounting period. To put this in context, erosion losses at the national scale have been estimated to be equivalent in magnitude to national electricity production or agricultural exports; approximately US$390 million per year, or 3.8 percent of GDP (Cohen, Brown and Shepherd, 2006, as cited in Turpie, Wilson, Hickinbotham, et al. (2025)). The Tana Basin, which originates in the agriculturally productive central highlands, carries sediment loads reaching downstream hydropower reservoirs at rates as high as 24,000 tonnes per day during rainy season surges; the Seven Forks cascade including Masinga and Kiambere dams already shows rapid siltation with a national average reservoir sedimentation rate of approximately 1.45 percent per year (Section 4.2, Turpie, Wilson, Hickinbotham, et al. (2025)). The sediment retention accounts provide the quantified physical and monetary baseline for cost-benefit analysis of catchment conservation investments protecting this hydropower infrastructure.

Ecosystem-based tourism. The value of ecosystem-based tourism, which is estimated as the resource rent attributable to natural ecosystem attractions, doubled from KSh 19,019 million (US$146 million) in 2002 to KSh 38,119 million (US$294 million) in 2018 (Table 4.8, Turpie, Wilson, Hickinbotham, et al. (2025)). This accounts for 84 percent of all attraction-based tourism value in Kenya, confirming that nature-based tourism, which consists mainly of wildlife viewing in the southern and central game areas, is the backbone of Kenya's tourism economy. In per-hectare terms, wildlife protected areas generate the highest average ecosystem tourism value, followed by forest reserves and community or private conservancies. Grassland and Acacia-Commiphora bushland together account for 82 percent of the national ecosystem-based tourism value (47 and 35 percent respectively), providing a direct spatial link between the condition of savanna ecosystems and the economic sustainability of the tourism sector. At the macro level, tourism accounts for 8.2 percent of Kenya's total economy (WTTC, 2020 as cited in Turpie, Wilson, Hickinbotham, et al. (2025)). In 2018, some two million people visited Kenya, spending KSh 193.8 billion (US$1.9 billion); of this, 63 percent of attraction-based expenditure was attributable to ecosystem-based activities. The accounts further document that wildlife populations declined from 31.9 percent of their 1977-1980 levels during 2011-2016 to an estimated 27.3 percent by 2018, projected to fall below 10 percent by 2050 under a business-as-usual scenario. This trajectory would erode the tourism asset value currently captured in the accounts. The cost of ecosystem loss and degradation in Kenya was estimated at approximately US$1.3 billion per year during 2001-2009 (Mulinge et al., 2015, as cited in Turpie, Wilson, Hickinbotham, et al. (2025)).

4.3.6 Forest Accounts (Thematic)

Kenya's forest accounts, published as a draft in January 2026, integrate physical asset accounts, supply and use tables for wood products, and an initial framework for monetary forest accounts (Kenya National Bureau of Statistics, 2026b). The accounts are anchored in the Kenya Forest Service's forest estate data and SLEEK-IMPRESS land cover time series, and are designed to inform Kenya's Vision 2030, NDC commitments, and REDD+ reporting. Forest cover was recorded at 3,591,486 ha (6.07 percent of national land area) in 2024, reduced from 3,934,279 ha (6.65 percent) in 1990. The six forest strata, comprising montane, western rainforest, coastal, mangrove, dryland, and public plantations, reflect Kenya's full ecological gradient from highland water towers to arid coastal belts. Public plantation forests managed by the Kenya Forest Service cover 152,790 hectares and are the primary source of commercial timber, poles, and charcoal. The Plantation Establishment and Livelihood Improvement Scheme (PELIS) allocates forest-adjacent land to communities for intercropping; profitability of PELIS arrangements was estimated at KSh 50,000 per hectare per year in the accounts, providing a measurable community benefit from forest management (Section 2e, Kenya National Bureau of Statistics (2026b)).

A distinguishing feature of Kenya's forest accounts is their integration with the Kenya Macroeconomic and Fiscal Model (KENMOD), which allows forest-related scenarios that include climate shocks to forest extent and the macroeconomic impact of forest restoration investments, to be translated directly into projections for GDP, fiscal revenues, and employment. This link to a macro-fiscal model is rare in the AFE region and positions Kenya to make concrete budget-level arguments for forest conservation that are directly comparable to those of any other economic sector. Monetary accounts for forest ecosystem services remain incomplete at this draft stage, with specific data gaps noted for informal timber flows from private forests and for non-timber forest products. The supply and use table structure for wood products is established, setting the framework for completing monetary valuation in the next edition. As in the ecosystem accounts, the most immediate valuation priorities are carbon sequestration (linked to Kenya's Climate Change Carbon Markets Regulations 2024), water regulation services from the water towers, and fuelwood valuation given that biomass accounts for more than 1.2 million TJ of Kenya's annual energy use; a flow that currently passes through the national accounts as essentially unpriced natural input.

4.4 Madagascar

Madagascar has compiled two SEEA Ecosystem Accounting accounts, covering ecosystem extent and ecosystem condition, based on an integrated landscape assessment covering the period 1992 to 2020 (Vogl et al., 2022). This work was developed by the World Bank's Biodiversity, Ecosystems and Landscapes team (BELA) and represents an experimental but spatially explicit ecosystem accounting exercise rather than a fully institutionalized SEEA-EA process. The accounts use a composite land degradation index derived from four sub-indicators, covering vegetation health (NDVI), carbon storage (net primary productivity), soil retention, and baseflow, to characterize ecosystem extent and condition trends across Madagascar's 59.2 million hectares at 30 to 300 metre resolution. No monetary ecosystem services accounts have been compiled. Nevertheless, the valuation methodology is described and referenced in the methodology supplement, but monetary results are presented separately and are not reproduced in the referenced document (Vogl et al., 2022). The accounts nonetheless provide a rich physical baseline with direct policy relevance for investments in nature-based solutions (NBS), watershed management, hydropower protection, and climate finance.

4.4.1 Ecosystem Extent and Condition Accounts (SEEA Ecosystem Accounting)

Madagascar's land cover was classified using the European Space Agency's Climate Change Initiative (ESA CCI) time series at 300 metre resolution from 1992 to 2020, reclassified into nine categories: tree cover, shrubland, grassland, cropland, built-up, bare or sparse vegetation, water body, wetland, and mangrove. The composite land degradation index, calculated as the product of trend slope and magnitude scores across the four sub-indicators, classified each pixel into one of seven degradation or improvement classes. At the national level, 35.6 percent of

Madagascar's land area (approximately 21.1 million hectares) showed degrading trends across the 29-year period. Results show a total of 7.4 million ha highly degrading (12.5 percent), 6.1 million ha moderately degrading (10.3 percent), and 7.6 million ha slightly degrading (12.8 percent). A further 32.7 percent of the territory (19.3 million ha) showed improving trends, while 31.7 percent (18.8 million ha) showed no statistically significant change (Vogl et al., 2022). These figures establish that over a third of the island's landscape experienced measurable ecosystem degradation over the reference period. This finding is directly relevant to Madagascar's obligations under the Kunming-Montreal Global Biodiversity Framework, its REDD+ commitments, and its Land Degradation Neutrality targets under the UNCCD.

The ecosystem condition accounts capture four services in physical units, providing the quantitative evidence base for investment planning.

Soil erosion. Under the 2020 business-as-usual baseline, total annual soil erosion estimated by the InVEST Sediment Delivery Ratio model was approximately 3.08 billion tonnes per year across the national territory (Vogl et al., 2022). This places Madagascar among the most erosion-prone landscapes in sub-Saharan Africa, consistent with documented lavaka erosion gullies in the central highlands. Of this, 590 million tonnes per year (19 percent) were generated within the highest-priority NBS investment zones. That is, areas where restoration would yield the greatest reductions in sediment load. Implementing integrated landscape management (ILM) in these top-priority zones would reduce soil erosion by approximately 81 percent within those areas, equivalent to roughly 479 million tonnes per year of avoided erosion. Agriculture productivity losses from high erosion (at rates exceeding 50-70 t/ha) have been estimated at approximately 8 percent of crop yields in affected regions, following Sartori et al. (2019) and Lal (1995), as applied in the landscape assessment. In rice cultivation, which is Madagascar's dominant crop, sediment-clogged irrigation channels have been documented to reduce paddy productivity by up to 10 percent in affected areas such as the Alaotra and Maroantsetra catchments (Carret and Loyer, 2003, as cited in Vogl et al. (2022)).

Sediment export and hydropower. Total annual sediment exported to waterways nationally was estimated at 27.2 million tonnes per year under the 2020 baseline (Vogl et al., 2022). The three main hydropower plants serving the Antananarivo interconnected grid;Andakaleka, Antelomita, and Mandraka; together provide approximately 90 percent of the grid's generation. Inadequate hydrological provision attributable to sedimentation and watershed degradation accounts for approximately 19 percent of generation losses relative to installed capacity. In the highest NBS priority zones, which overlap largely with the upstream catchments of these three plants, sediment export was 9.7 million tonnes per year, and ILM implementation could reduce sediment export by approximately 81 percent in those areas. This potential translates directly into avoided costs of dredging (estimated at US$4.5-9.4 per tonne as cited in Vogl et al. (2022)) and avoided generation losses, which are values that have been quantified in the economic analysis accompanying the landscape assessment but are not reproduced in the methodology supplement.

Surface runoff. Total annual quick flow (surface runoff) under the 2020 baseline was estimated at approximately 39.6 billion m^3 per year nationally (Vogl et al., 2022). In the highest-priority NBS investment zones, quick flow was approximately 11.1 billion m^3 per year. ILM implementation in these zones could reduce quick flow by 73.5 percent within them, which is equivalent to approximately 8.2 billion m^3 per year of reduced surface runoff, and directly translates into lower flood risk, higher dry-season baseflow, and improved water availability for agriculture and urban supply. The overall national quick flow reduction achievable through full ILM implementation was estimated at 26.3 percent, underscoring the scale of potential water regulation benefits.

Carbon storage. Total carbon stored in Madagascar's terrestrial vegetation and soils under the 2020 baseline was estimated at approximately 6.25 billion tonnes of carbon (Vogl et al., 2022). Of this, 1.96 billion tonnes (31 percent) are located within the highest-priority NBS zones, meaning that these areas, which represent only 7.5 percent of national territory, contain nearly a third of the country's carbon stock. This extreme concentration of carbon in high-priority, high-vulnerability areas is a critical finding for climate finance, because it indicates that the landscapes most at risk of degradation are also those with the highest potential carbon credit value. The landscape assessment separately estimated that the average annual reduction in carbon absorption (net primary productivity) due to land degradation was 1.56 million tonnes of CO per year over the analysis period (Section 2.1, Vogl et al. (2022)). At the Atiala Atsinanana REDD+ program reference price of US$5 per tonne CO, this annual absorption loss alone represents a foregone revenue of approximately US$7.8 million per year, while at the social cost of carbon applied in the study (US$38.1 per tonne CO in 2015, rising 3 percent annually), the annual cost reaches approximately US$59 million per year. These estimates are conservative, as they address only the degradation of carbon absorption capacity, not the direct loss of carbon stocks from forest clearing.

Biodiversity threat. The accounts also incorporate an IUCN STAR biodiversity threat layer, which indicates that 300,753 ha (0.51 percent of national area) are classified in the two highest-threat classes (0.1 and 0.2), with a further 1.49 million ha in moderate-high threat classes (0.3-0.4). Madagascar's exceptional biodiversity significance, with approximately 90 percent endemism across many taxonomic groups, means that even small areas of habitat loss carry globally disproportionate consequences. The overlap between high biodiversity threat areas and high NBS priority zones provides the spatial evidence base for arguing that watershed restoration investments in Madagascar generate co-benefits for global biodiversity conservation that are not captured in any current monetary account.

Data gaps and next steps. The accounts do not yet include monetary valuation, and there are no physical supply and use tables disaggregating ecosystem service flows by sector or user group. The economic analysis of hydropower losses, agricultural productivity costs, and carbon opportunity costs was computed as part of the broader BELA landscape assessment (2022-2023) but is presented in a companion report not available in the current data repository. The immediate priorities for completing Madagascar's ecosystem accounts include publishing the full economic results from the landscape assessment to make the monetary estimates of land degradation costs formally available; disaggregating ecosystem service flows by basin and administrative region using the 22-region database already compiled in the supporting Excel data; extending the analysis beyond 2020 with updated land cover data; and institutionalizing the accounts within INSTAT (Madagascar's national statistics institute) to enable regular production.

4.5 Mauritius

Mauritius has compiled two SEEA Central Framework water accounts, consisting of physical supply and use tables for water and physical asset accounts for water, published as the fifth edition of the *Water Account, Mauritius* report by Statistics Mauritius in December 2024 (Statistics Mauritius, 2024). The series covers the years 2021 to 2022 with a time series of selected indicators extending back to 2013. The accounts are based on the SEEA-Water framework and the International Recommendations for Water Statistics (IRWS), and are produced in collaboration with the Water Resources Unit (Ministry of Energy and Public Utilities), the Central Water Authority (CWA), and the Waste Management Authority (WMA). The fact that this is the fifth edition shows that Mauritius has the most institutionalized and regularly produced water accounting system in the AFE region; a model for countries earlier in the SEEA implementation path.

4.5.1 Physical Supply and Use Tables for Water (SEEA-CF)

Water availability and stress. Total Renewable Water Resources (TRWR) were 2,643 million m^3 (Mm^3) in 2021 and 2,874 Mm^3 in 2022, driven by annual rainfall of 2,025 mm and 2,201 mm respectively, both close to the long-term average of 2,018 mm (1991-2020) for the island (Statistics Mauritius, 2024). Of each year's precipitation, 60 percent becomes surface runoff, 30 percent is lost to evapotranspiration, and 10 percent recharges groundwater. Total freshwater abstracted from the environment for use in the economy was 604 Mm^3 in 2021 and 632 Mm^3 in 2022 (excluding hydropower). As a proportion of TRWR, abstraction was 22.9 percent in 2021 and 22.0 percent in 2022, which is below the 25 percent threshold at which a territory is classified as water-stressed by UN-Water. Mauritius therefore remains in the "no stress" class, though the margin is narrow and total abstraction has been rising steadily from around 591 Mm^3 in 2017. TRWR per capita stood at 2,163 m^3 per inhabitant in 2021 and 2,360 m^3 per inhabitant in 2022 (Statistics Mauritius, 2024).

Abstraction by sector. The largest single user in physical terms is the hydropower sector, which abstracted 364 Mm^3 (37.6 percent of total) in 2021 and 431 Mm^3 (40.5 percent) in 2022, with the caveat that hydropower is a non-consumptive use and virtually all of this water is returned to the environment (Statistics Mauritius, 2024). Agriculture is the second-largest abstractor, withdrawing 300 Mm^3 in 2021 and 302 Mm^3 in 2022 (28.4 percent of total). The CWA, the public water utility, abstracted 295 Mm^3 in 2021 and 320 Mm^3 in 2022 for treatment and distribution. Manufacturing and services abstracted only 9-10 Mm^3, reflecting Mauritius's service-dominated economy. Across the full ten-year series (2013-2022), the hydropower sector shows the highest growth in abstraction, rising from 280 Mm^3 to 431 Mm^3 (+54 percent), coinciding with Mauritius's renewable energy expansion. Hydropower's share of electricity generation grew from 3.3 percent in 2013 to 4.2 percent in 2022 (Statistics Mauritius, 2024).

Water utilisation and consumption. Total water utilised by all sectors combined was 1,129 Mm^3 in 2021 and 1,224 Mm^3 in 2022; figures that include both direct environmental abstraction and water received from the CWA utility network. Of the 1,224 Mm^3 utilised in 2022, hydropower accounted for 35.2 percent (431 Mm^3), the CWA for 26.1 percent (320 Mm^3), agriculture for 24.9 percent (304 Mm^3), households for 7.1 percent (87 Mm^3), and manufacturing and services for the remaining 2.8 percent (34 Mm^3) (Statistics Mauritius, 2024). Net water consumption, which refers to the volume that entered the economy but did not return to either inland water resources or the sea, was 226 Mm^3 in 2021 and 229 Mm^3 in 2022, equal to 21.5 percent of total abstraction in 2022. Agriculture dominates consumption, accounting for 204 Mm^3 (89.1 percent of net consumption) in 2022, as irrigation water is incorporated into crops or evapotranspired. Households consumed 13 Mm^3 (5.7 percent) and manufacturing 12 Mm^3 (5.2 percent).

Unaccounted-for water. The most policy-significant indicator in the supply and use tables is the rate of Unaccounted-for Water (UFW) in the CWA distribution network. In 2022, the CWA abstracted 320 Mm^3 but lost 207 Mm^3 to distribution leakages, which is a UFW rate of 64.6 percent (Statistics Mauritius, 2024). This means that nearly two-thirds of treated potable water abstracted by the utility is lost before reaching end users. The trend is worsening, because UFW was 56.4 percent in 2013, but rised to 64.6 percent by 2022. In absolute volume, this represents treated water equivalent to the entire annual agricultural abstraction, disposed of as distribution losses. Against this, households received an average of only 197 litres per capita per day through the pipe network in 2022 (up from 165 L/day in 2013), despite 99.8 percent of the population having access to safely managed drinking water services in 2022 (2022 Housing and Population Census, cited in Annex 1, Statistics Mauritius (2024)). The gap between abstraction for drinking water per capita per day (719 L in 2022) and water actually received by households (197 L/day) is almost entirely explained by the UFW rate. Quantifying the economic cost of UFW, in terms of treatment cost, pumping energy, and foregone revenue, would be a natural first application of monetary water accounts in Mauritius, and a compelling policy argument for infrastructure investment.

4.5.2 Physical Asset Accounts for Water (SEEA-CF)

The physical asset accounts for water resources link environmental water stocks to the flows abstracted and returned by the economy (Tables 5-6, Statistics Mauritius (2024)). In 2022, total additions to the water resource stock were 5,290 Mm^3, an increase of 9.4 percent over the 4,834 Mm^3 recorded in 2021, reflecting higher rainfall. Surface water additions (reservoirs, lakes, rivers, and streams) were 2,894 Mm^3, groundwater additions 754 Mm^3, and soil water 1,642 Mm^3. Total reductions from the stock in 2022 were also 5,290 Mm^3, comprising: sea outflows of 2,585 Mm^3, evapotranspiration of 1,231 Mm^3, abstraction (including hydropower) of 1,063 Mm^3, and outflows to other inland water resources of 411 Mm^3. Of total available surface water in 2022, 31.3 percent was abstracted; of groundwater, 20.8 percent was withdrawn. Both figures are well within sustainable limits in aggregate, but the asset accounts reveal that surface water use is approaching the level at which inter-annual variability in rainfall, which ranged from 1,896 mm (2016) to 2,816 mm (2018) in the ten-year series, could create water stress. The accounts note that in the absence of stock data at the beginning and end of each year, the balance of additions and reductions is calculated as a simplified equilibrium assumption (Section 4, Statistics Mauritius (2024)), indicating a key data gap for future account development.

Data gaps and next steps. Mauritius's water accounts are physical-only and no monetary water accounts have been compiled. The most obvious extension is the monetisation of UFW losses, which would quantify the annual financial cost to the utility of distribution infrastructure deterioration. A second priority is developing monetary supply and use tables that assign the economic value of water abstraction to each sector, with particular emphasis on agriculture, which consumes the largest net volume, and hydropower, whose share of the electricity mix is growing. The accounts also do not yet extend to the outer islands of Mauritius (Rodrigues, Agaléga, and St Brandon), all of which face distinct water availability challenges given their remote location and limited groundwater resources.

4.6 Rwanda

Rwanda has compiled five SEEA accounts spanning three domains, consisting of water (physical and monetary supply and use tables), land (physical asset accounts), and ecosystems (condition and physical services), making it one of the more advanced NCA programmes in the AFE region. The accounts were produced by the National Institute of Statistics of Rwanda (NISR) in collaboration with the Ministry of Environment and MINECOFIN, with World Bank WAVES technical support, and were published as Version 1.0 reports in 2018 and 2019 (Government of Rwanda, 2018, 2019b, 2019a). Rwanda's accounts are notable for including monetary supply and use tables for water; one of the very few such accounts in Africa; and for explicitly linking water use to GDP through water productivity indicators, providing a direct bridge between natural capital and macroeconomic planning. Rwanda's total wealth is estimated at US$21,619 per capita (2014), of which natural capital accounts for US$6,650 (31 percent), and according to discounted returns analysis the future value of natural capital is approximately four times that of produced assets (World Bank / Lange et al. 2018, cited in Figure 1, Government of Rwanda (2019a)).

4.6.1 Physical and Monetary Supply and Use Tables for Water (SEEA-CF)

Water resources and stress. Rwanda's Total Renewable Water Resources (TRWR) fluctuated between 10,329 and 11,925 million m^3 per year over the 2012-2015 period (Table 9, Government of Rwanda (2019b)), yielding per-capita TRWR availability of between 941 and 1,112 m^3 per year, situated close to the internationally recognised water scarcity threshold of 1,000 m^3 per capita. Water stress, measured as total freshwater withdrawn as a proportion of available resources (SDG indicator 6.4.2), was 7.2 percent in 2012, rising to 8.3 percent in 2013, 7.4 percent in 2014, and 8.1 percent in 2015 (Table 10b, Government of Rwanda (2019b)), situated below the 25 percent stress threshold, but with a rapidly growing demand trajectory. The National Water Resources Master Plan (RNRA 2015) projects demand rising from approximately 120 million m^3 per year to 3.37 billion m^3 per year by 2040, driven primarily by irrigation expansion (Table 7, Government of Rwanda (2019b)), which represents a ten-fold increase that would push abstraction to approximately 30 percent of TRWR, crossing into the stressed category.

Sectoral abstraction and the agriculture dominance. Agriculture accounts for over 96 percent of total water abstraction in Rwanda (Figure 7, Government of Rwanda (2019b)), principally through large-scale irrigation diversions from rivers and through soil water use. Total water abstraction including soil water was approximately 14,039 million m^3 in 2012, with agriculture drawing 13,563 million m^3 (Table 1, Government of Rwanda (2019b)). Net freshwater withdrawal from surface and groundwater totalled 656 million m^3 in 2012, rising to 716 million m^3 in 2015. Hydropower abstraction grew approximately 20 percent between 2012 and 2014, and household abstraction grew 134 percent over the same period (Figure 1, Government of Rwanda (2019b)), reflecting both population growth and expanding urban water services. Despite agriculture's overwhelming share of water use, it generates only 30 percent of GDP, while agriculture water productivity, situated at 118 Rwf per m^3 in 2015, is the lowest of any sector, compared to 523 Rwf/m^3 in manufacturing, 6,236 Rwf/m^3 in mining, and between 9,300 and 11,600 Rwf/m^3 in services (Table 11 and Figure 11, Government of Rwanda (2019b)). This mismatch between high water consumption and low economic return per unit of water is the central message of the monetary accounts.

Monetary water productivity and macroeconomic linkage. The monetary supply and use tables, which are the only such accounts produced for water in the documents reviewed for the AFE region, estimate national water use efficiency at 4,762 Rwf per m^3 of water used in 2015 (at constant 2014 prices), up from approximately 4,500 Rwf/m^3 in 2012, This is an improvement of approximately 13 percent (Figures 13-14, Government of Rwanda (2019b)). It reflects a period of relative decoupling, in which Rwanda's GDP grew 25 percent between 2012 and 2015, while total water use grew only 10 percent (Figure 15, Government of Rwanda (2019b)). For comparison, Rwanda's 2012 water productivity was US$23.4 per m^3 using FAO/AQUASTAT methodology (SDG 6.4.1, cited in Government of Rwanda (2019b)). In absolute macroeconomic terms, water sector gross value added (ISIC E, water and waste management) was approximately 31 billion Rwf in 2015 and 33 billion Rwf in 2016 (in constant 2014 prices), a relatively small share of GDP of 5,951 billion Rwf in 2015 (GDP tables, Government of Rwanda (2019b)). The water accounts also identify that climate-related impacts on water resources could cost Rwanda approximately 1 percent of GDP per year by 2030 (MINIRENA 2011b, cited by Government of Rwanda (2019b)), providing a quantified argument for investment in watershed protection.

Water utility economics and distribution losses. Rwanda's Water and Sanitation Corporation (WASAC) provides potable water to approximately 2.5 million people, out of a total of 4.3 million (37.3 percent of the population) served by all utilities (Table 12, Government of Rwanda (2019b)). WASAC's annual abstraction capacity is approximately 42 million m³ per year. Tariffs are differentiated by use type, where public taps (kiosk/household connections) are charged 323 Rwf/m³, residential block rates range from 340 to 877 Rwf/m³, and industrial users pay 736 Rwf/m³ (Tables 13-14, Government of Rwanda (2019b)). However, WASAC achieves only approximately 75 percent cost recovery; the break-even tariff is approximately 700 Rwf/m³, and an annual government subsidy of approximately 5.2 billion Rwf is required. Non-Revenue Water (NRW), which is water produced but not billed, accounts for approximately 41 percent of WASAC output, equal to 16.5 million m³ per year and US$8.7 million in foregone revenue (Karamage et al. 2016, cited in Government of Rwanda (2019b)). Halving NRW would free up an additional 8 million m³ per year, equivalent to supply for 42,000 additional households or irrigation for 660 additional hectares, and would add approximately US$4 million per year in revenue. The asset accounts show that reservoir storage in Rwanda increased by 63 percent between 1990 and 2015, while the sediment load in those reservoirs grew by 123 percent over the same period, and by 39 percent in just 2010-2015 alone (Table 6 and Text Box 2, Government of Rwanda (2019b)), indicating that sedimentation from watershed erosion is already materially reducing storage capacity.

4.6.2 Physical Asset Accounts for Land (SEEA-CF)

Rwanda's land accounts, published in March 2018, cover two accounting periods (2014 and 2015) using the Rwanda Land Administration and Information System (LAIS) for registered parcels and RCMRD satellite land cover for 1990, 2000, 2010, and 2015 (Government of Rwanda (2018)). The LAIS area covered is 2,069,548 ha, representing registered land; the full national territory is approximately 2,488,123 ha (Table 10, Government of Rwanda (2018)). Agriculture is the dominant land use at 60.6 percent of registered area (1,242,363 ha) as of 2014, with net annual additions of 8,818 ha in 2014 (Table 3, Government of Rwanda (2018)). Forestry covers 9.3 percent (190,612 ha) and livestock 5.7 percent (120,445 ha).

Land fragmentation. The most striking structural feature of Rwanda's land tenure is extreme fragmentation, because the national average parcel size is 0.184 ha (2014), while approximately 70 percent of parcels are smaller than 0.2 ha, and only 3 percent exceed 1 ha (Table 11, Government of Rwanda (2018)). With Rwanda's population density of approximately 414 persons per km² (2012) and more than 60 percent of households farming less than 0.7 ha, this fragmentation poses a structural constraint on both agricultural productivity and sustainable land management.

Land cover change. The RCMRD time series reveals a dramatic transformation since 1990 in which annual cropland expanded from 613,527 ha (24.2 percent of national area) in 1990 to 1,310,704 ha (51.7 percent) by 2015; a 113.6 percent increase (Table 13, Government of Rwanda (2018)). Sparse forest declined from 942,347 ha (37.1 percent) to 269,210 ha (10.6 percent); a loss of 673,137 ha, with 306,663 ha lost in just the five years from 2010 to 2015. Total forest cover (dense, moderate, and sparse combined) fell from 43.0 percent in 1990 to 17.0 percent by 2015. Dense forest reversed this trend, increasing from 84,151 ha to 115,762 ha (+37.6 percent) due to plantation programmes and national park protection, pushing Rwanda's combined forest and shrubland cover to 30.15 percent by 2015, meeting the Vision 2020 forest target of 30 percent (Table 14a, Government of Rwanda (2018)). Wetlands declined from approximately 110,000 ha (4.3 percent) to 88,773 ha (3.5 percent) between 1990 and 2015, with about 14,000 ha lost in the 2010-2015 window alone (p. 35, Government of Rwanda (2018)).

Monetary land values and transactions. The land accounts include an initial monetary dimension based on land transaction data. In 2014, 15,520 registered transactions covered 4,020 ha at a total value of 141,825 million Rwf, with a national average of 35 million Rwf per hectare (Table 15, Government of Rwanda (2018)). Kigali City transactions accounted for 62 percent of total transaction value despite a small share of area, with average land values of 130 million Rwf per hectare; more than three times the national average (Table 17, Government of Rwanda (2018)). Land market data, while not yet integrated into a full asset account balance, establish that Rwanda's land asset base is substantial and increasingly monetized. Soil erosion costs have been estimated at US$34 million per year, equivalent to approximately 2 percent of GDP (REMA 2009, cited in Government of Rwanda (2018)), providing direct motivation for accounting for the degradation of the land asset.

4.6.3 Ecosystem Condition and Physical Ecosystem Services Accounts (SEEA-EA)

Rwanda's ecosystem accounts, published in November 2019, cover the years 1990, 2000, 2010, and 2015 and use the InVEST modelling platform to estimate four physical ecosystem services: soil erosion, sediment export, water yield, and quickflow (surface runoff) and baseflow (Government of Rwanda, 2019a). No monetary valuation of ecosystem services is included in these accounts.

Soil erosion. Potential soil erosion nationally increased from 102.5 million tonnes per year in 1990 to 158.2 million tonnes per year in 2015; a 54 percent increase (Table 1, Government of Rwanda (2019a)). The 2015 national average of 62 tonnes per hectare per year, which is approximately 13 tonnes per capita per year, is among the highest in Sub-Saharan Africa. The Northern Province is the most severely affected at 109 t/ha/yr, the Mukungwa catchment followed at 129 t/ha/yr (Figure 6, Government of Rwanda (2019a)), and the Western Province averages 94 t/ha/yr. The eastern province is comparatively low at 24 t/ha/yr. The Upper Nyabarongo and Lower Nyabarongo catchments together account for approximately 37 percent of national erosion loads (Figure 5, Government of Rwanda (2019a)). These rates far exceed the natural regeneration capacity of soils and are directly linked to the conversion of 37 percent of land from sparse forest to annual cropland between 1990 and 2015, with over 90 percent of cultivated land on slopes greater than 10 percent (Section 2.1, Government of Rwanda (2019a)).

Sediment export. Sediment reaching waterways increased by 123 percent between 1990 and 2015, from 6.3 million tonnes per year to 14.0 million tonnes per year (Table 2, Government of Rwanda (2019a)), with the 2015 national average of 5.5 t/ha/yr. The Mukungwa catchment exported 11.39 t/ha/yr (2,082,366 tonnes total), the highest rate nationally. This growing sediment load is directly correlated with the 123 percent increase in sediment in Rwanda's reservoirs documented in the water accounts and provides the biophysical basis for quantifying the economic cost to the hydropower and potable water sectors.

Water yield and hydrological balance. Total water yield from Rwanda's landscape increased modestly from 7.9 to 8.2 billion m³ per year between 1990 and 2015 (+4.1 percent, Table 3, Government of Rwanda (2019a)). However, beneath this aggregate trend lies shift in the composition of water yield. Quickflow (surface runoff) increased by 35 percent, from 2.4 to 3.2 billion m³ per year, while baseflow (groundwater-fed dry-season flow) declined by 11.4 percent, from 5.1 to 4.5 billion m³ per year (Tables 4 and 6, Government of Rwanda (2019a)). By 2015, quickflow accounted for 39 percent of total water yield and baseflow 54 percent, compared to 30 and 64 percent respectively in 1990. This hydrological shift, which results in more peak flow and less sustained dry-season flow, has direct consequences for hydropower reliability, municipal water supply, and agricultural irrigation. Baseflow declined in every district between 1990 and 2015 (Annex 6, Government of Rwanda (2019a)), with the greatest losses in peri-urban Kigali districts (Kicukiro −32 percent, Ngoma −29 percent, Nyarugenge −27 percent). Districts in the Lake Kivu and Mukungwa catchments are experiencing quickflow increases of between 78 and 84 percent, compounding flood risk. The Upper Akagera catchment, which is the source of the Nile in Rwanda, has already experienced a 24 percent decline in per-hectare baseflow (from 398 to 300 m³/ha/yr), threatening water security for the semi-arid East (Figure 13, Government of Rwanda (2019a)).

4.7 Uganda (UGA)

Uganda has compiled six SEEA accounts across two published programmes: physical water supply and use tables from the Uganda Bureau of Statistics (UBOS) water accounting series (2015-2024), and a set of five forest-based accounts, consisting of monetary timber asset accounts, ecosystem extent, physical and monetary ecosystem services, and ecosystem asset accounts, from the Uganda Wood Asset and Forest Resources Accounts (Uganda Bureau of Statistics, 2020). Together, these accounts document a country where forest resources underpin between 4 and 8 percent of GDP and 78 percent of national energy supply (Uganda Bureau of Statistics, 2020), yet forest cover fell by 60 percent between 1990 and 2015. Uganda's water resources are enormous, with total abstraction reaching approximately 499 billion m³ per year including rainfed agriculture, but institutional use efficiency is declining and distribution losses remain high.

4.7.1 Physical Supply and Use Tables for Water (SEEA-CF)

Uganda's water accounts, now in their tenth iteration (2015-2024), are among the most systematically maintained in the AFE region. The 2024 edition (Uganda Bureau of Statistics, 2025) reports total gross water abstraction of 499,039 million m³, of which rainfall for rainfed agriculture and natural vegetation dominates (278,872 million m^3, 56 percent), followed by surface water abstractions (219,916 million m^3, −3.2 percent from 2023), and groundwater (252 million m^3, +6.1 percent; Table 2, Uganda Bureau of Statistics (2025)). Total consumptive use is 37,957 million m^3 (+1.2 percent), while return flows constitute 423,022 million m^3 or 55.9 percent of abstracted water (Table 1, Uganda Bureau of Statistics (2025)).

Sectoral use and agriculture's dominance. Agriculture accounts for 55.4 percent of total abstraction, and 96.4 percent of all consumptive use, where rainfed crops alone account for 45.4 percent of consumption, livestock 34.5 percent, and managed forestry 19.6 percent (Table 6, Uganda Bureau of Statistics (2025)). Uganda's agriculture contributes approximately 24 percent of GDP and 42 percent of export earnings (FY 2023/24), yet more than 96 percent of cultivated area is rainfed; irrigated land represents only 0.5 percent of the nation's agronomic potential, which is one of the lowest uptake rates among regional peers (Section 1.1, Uganda Bureau of Statistics (2025)). Industry accounts for 44.6 percent of abstraction, overwhelmingly driven by electricity generation, given that hydropower abstracts 218,326 million m^3 per year. An emerging pressure point is crude oil and mining, whose abstraction increased by 113.9 percent from 2023 to 2024, from 164,676 to 352,188 thousand m^3, evidencing demands of Uganda's oil development programme (Table 3, Uganda Bureau of Statistics (2025)).

Water productivity and efficiency. The accounts include a monetary dimension by linking water use to GDP value added. National water productivity reached UGX 5,557 per m^3 consumed in 2024, up 8.2 percent from UGX 5,137 in 2023; and more than 84 percent above the 2015 baseline of UGX 3,014 per m^3 (Figure 4 and Table 7, Uganda Bureau of Statistics (2025)). This positive trend is moderated by a decline in water use efficiency (value added per m^3 of water used excluding precipitation), which fell 0.8 percent to UGX 99,865 per m^3 (approximately US$27/m^3) in 2024 (Table 8, Uganda Bureau of Statistics (2025)). Agriculture's water productivity is the lowest among productive sectors (UGX 1,458 per m^3), while the services sector achieves the highest water use efficiency (UGX 364,299 per m^3). Distribution losses in the supply chain remain substantial at approximately 34.8 percent (Table 9, Uganda Bureau of Statistics (2025)). Household-level water availability is low and per-capita annual consumption is 825 litres and per-household daily consumption is 27 litres, which is well below the 50-litre per person per day benchmark for basic needs (Figure 7, Uganda Bureau of Statistics (2025)).

4.7.2 Monetary Asset Accounts for Timber Resources and Ecosystem Accounts (SEEA-CF and SEEA-EA)

Uganda's forest accounts, produced by UBOS under the WAVES programme and published in June 2020, compile data for 1990, 2000, 2005, 2010, and 2015. They document one of the most acute deforestation crises in the region.

Forest extent and loss. Uganda's forest cover fell from 4,933,762 ha in 1990 (approximately 24 percent of national territory) to 1,951,662 ha in 2015; a loss of 2,982,100 ha, or roughly

119,200 ha per year for 25 years (Table 3.1, Figure 3.1, Uganda Bureau of Statistics (2020)). The dominant conversion is woodlands, showing that woodland area fell from 3,974,523 ha in 1990 to 1,212,951 ha in 2015 (−69 percent), with 1,167,265 ha converted to smallholder farms, 885,862 ha to grassland, and 503,483 ha to bush (Table 3.2, Uganda Bureau of Statistics (2020)). By contrast, planted forests expanded markedly, where broadleaved plantations grew from 18,682 ha to 44,237 ha (+137 percent) and coniferous plantations from 16,384 ha to 63,486 ha (+287 percent), but at a scale insufficient to offset woodland losses. Plantations offset only approximately 6 percent of the net stock decline (Section 7.1, Uganda Bureau of Statistics (2020)). Poverty fell in tandem with deforestation, showing that the headcount ratio declined from 56.4 percent in 1990 to 21.4 percent in 2015. This highlights the difficult trade-off between forest protection and rural livelihoods expansion (Figures 6.3-6.5, Uganda Bureau of Statistics (2020)).

Monetary forest land value. The monetary ecosystem asset accounts value forest land using discounted land prices derived from government Resettlement Assessment Plans, disaggregated by sub-region and accounting period. Total national forest land value was estimated at US$27.6 billion in 1990, peaked at US$62.9 billion in 2005, and stood at US$34.9 billion in 2015, showing a nominal increase of 26.7 percent over 25 years despite the 60 percent loss in forest area (Figures 3.3-3.4, Uganda Bureau of Statistics (2020)). The paradox of rising total value alongside area loss is explained by rapidly escalating per-hectare land prices; especially in the Central region, where forest land averaged US$129,700 per hectare in 2015 and the Central 1 sub-region alone accounted for 61 percent of national forest land value. The national average rose from US$7,800 per hectare in 1990 to US$19,700 per hectare in 2015 (Figure 3.4, Uganda Bureau of Statistics (2020)), reflecting both urbanisation pressure and increasing land scarcity.

Monetary wood asset accounts. The national aggregate monetary wood stock, valued at stumpage prices, declined from US$9.1 billion in 1990 to US$7.4 billion in 2015 (−19 percent), with available wood stock (for supply) falling more sharply from US$2.5 billion in 1990 to US$1.3 billion in 2015 (Table 4.4, Figures 4.7 and 4.1, Uganda Bureau of Statistics (2020)). Physical wood stock fell from 355.5 million tonnes to 197.1 million tonnes (−45 percent) over the same period (Table 4.1, Uganda Bureau of Statistics (2020)), with wood available for supply declining from 224.6 Mt in 2000 to 91.5 Mt in 2015 (−53 percent). The accounts project that sustainable forest wood supply will be fully depleted by approximately 2025 under business-as-usual conditions, with a supply deficit growing from 1.6 Mt in 2000 to 35 Mt in 2015 and projected 105 Mt by 2040 (Figure 6.8, Section 7.1, Uganda Bureau of Statistics (2020)).

Monetary wood product supply and energy dependency. Wood product supply values by category in 2015 include sawn timber (US$111.7 million), poles (US$77.7 million), household firewood (US$45.3 million), commercial firewood (US$8.5 million), industrial firewood (US$6.8 million), and charcoal (US$20.6 million); a total of US$270.6 million in 2015, compared to US$92.7 million in 1990 (Figure 6.11, Uganda Bureau of Statistics (2020)). The physical volume of charcoal production grew from 2,396 thousand tonnes in 1990 to 11,962 thousand tonnes in 2015, consuming approximately 95.7 million tonnes of wood per year by 2015 (69 percent of total wood input), at a charcoal kiln efficiency of just 12.5 percent, or roughly 8 kg of wood per 1 kg of charcoal (Section 5.2, Uganda Bureau of Statistics (2020)). Woody biomass provides 78 percent of Uganda's national energy supply (MEMD 2016, cited in Uganda Bureau of Statistics (2020)), making forest depletion simultaneously an energy security crisis. Forestry's contribution to GDP is estimated at 4 percent (UBOS 2018) and up to 8 percent when informal uses are included (NEMA 2011, cited in Uganda Bureau of Statistics (2020)).

Monetary ecosystem services. Uganda's accounts include monetary values for non-wood forest products (NWFPs) as part of the monetary ecosystem services accounts. In 2015, shea butter (from shea pods) was valued at UGX 14,553 million and *Prunus africana* bark at UGX 265,195 million, with exports of *Prunus africana* alone generating UGX 12,628 million in foreign exchange (Section 7.2, Uganda Bureau of Statistics (2020)). Combined NWFP value reached approximately US$86.7 million in 2015. Uganda has historically had a trade deficit in wood and wood products, but this has narrowed from UGX 63 billion in 2000 to UGX 7 billion in 2015, partly reflecting growth in domestic processing (Section 7.1, Uganda Bureau of Statistics (2020)). The accounts note that approximately 4.7 percent of the monetary value of wood available for supply was captured in the formal economy in 1990 (Section 5.3.2, Uganda Bureau of Statistics (2020)), indicating very large informal and subsistence flows that are accounted for but not commercially intermediated.

4.8 South Africa (ZAF)

South Africa operates the most comprehensive natural capital accounting programme in the AFE region. Statistics South Africa (Stats SA) has published seven volumes in its Natural Capital Series, with accounts covering water (including the unique distinction of water emission accounts), energy, land, ecosystem extent, ecosystem services, and protected areas. South Africa's seven compiled accounts span 1990-2022 and collectively demonstrate a country in which physical natural capital trends are mixed. For example, natural vegetation is largely stable nationally but under acute pressure in water-producing highlands, while energy accounts reveal extreme coal dependency alongside accelerating oil import exposure. Biodiversity-based tourism has been quantified and linked to GDP, providing the only monetary ecosystem services estimate in the AFE region outside Kenya.

4.8.1 Physical Supply and Use Tables for Water, Physical Asset Accounts for Water, and Water Emission Accounts (SEEA-CF)

South Africa's water accounts focus on the 22 designated Strategic Water Source Areas (SWSAs), which are the mountain and highland catchments that supply the nation's water despite covering only 8.2 percent of national land area (Statistics South Africa, 2023). The SWSAs (10,020,720 ha in South Africa plus 2,389,969 ha in Lesotho and Eswatini) supply approximately 50 percent of the population, roughly two-thirds of national economic activity, 70 percent of irrigated agriculture, and more than 90 percent of urban water users (Section 1, Statistics South Africa (2023)). Seven SWSAs are transboundary and four extend into Lesotho and three into Eswatini, making SWSA management inherently a regional governance issue.

Land cover change in strategic catchments. Between 1990 and 2020, natural and semi-natural land in the SWSAs declined from 70.6 percent to 68.7 percent of SWSA area, a net loss of 181,141 ha (−2.6 percent; Tables 9-10, Statistics South Africa (2023)). Timber plantations expanded by 71,847 ha (+5.4 percent), urban extent by 65,277 ha (+16.5 percent), and mining by 51.9 percent. The SWSAs are substantially more altered than the national average. Intensively modified land covers 28.8 percent of SWSAs versus 16.2 percent nationally (Figure 10, Statistics South Africa (2023)), driven by the high concentration of timber plantations (13.9 percent of SWSAs vs. 1.7 percent nationally). Population density in the SWSAs averages 140 people per km^2, more than three times the national average of 42 per km^2, meaning they are simultaneously the source of water and sites of intense human pressure (Table 8, Statistics South Africa (2023)).

Ecological integrity threshold. A critical ecological finding is that three SWSAs had fallen below the 60 percent natural land cover threshold considered necessary for functional water production, consisting of Upper Usutu (40.9 percent natural in 2020), Table Mountain (50.4 percent), and Mpumalanga Drakensberg (51.3 percent). Others, including Upper Vaal (60.1 percent) and Southern Drakensberg (60.4 percent), are at the threshold (Figure 14, Statistics South Africa (2023)). Southern Drakensberg lost an additional 56,011 ha to intensive use over the period, and Enkangala lost 51,186 ha (+35.7 percent). These catchments are responsible for regulating water for Gauteng, South Africa's economic core. Protected area coverage in the SWSAs reached 18.9 percent by 2020 (from 13.9 percent), substantially above the national mainland average of 9.2 percent, but unevenly distributed, because Swartberg is 76.5 percent protected while Eastern Cape Drakensberg protected area is only 1.1 percent (Table 15, Statistics South Africa (2023)).

Biome composition and hydrology. Grassland dominates SWSA land cover (63.4 percent, 6,351,848 ha), followed by Fynbos (20.0 percent) and Savanna (12.5 percent; Table 5, Statistics South Africa (2023)). Forests cover 2.2 percent but account for 47.8 percent of the national forest biome within SWSAs, underscoring the water-provisioning role of afforested landscapes. The accounts note that no single TRWR or sector abstraction table is reported within the SWSA accounts, because these are addressed in the parallel national water accounts.

4.8.2 Physical Supply and Use Tables for Energy (SEEA-CF)

South Africa's physical energy flow accounts, covering 2015-2022 and published as Natural Capital Series 7, are the most detailed energy accounts in the AFE region (Statistics South Africa, 2025). Total domestic natural energy inputs declined slightly from 7,170.7 PJ in 2015 to 6,518.6 PJ in 2022, as renewable inputs grew from 0.8 to 1.4 percent of the total while coal, still 72.2 percent of domestic supply, contracted. Total national energy supply (including imports and transformation) was 20,910.6 PJ in 2022 (Table 4, Statistics South Africa (2025)).

Coal dominance and export role. Coal extraction provides approximately 5,454 PJ from mining and 54 PJ from manufacturing in 2022, and electricity generation produces a further 896 PJ from coal transformation. Exports totalled 2,046.9 PJ in 2022, of which coal accounted for 90.4 percent, making South Africa a major global coal exporter even as domestic energy transformation is contested (Tables 19-20, Statistics South Africa (2025)). Industrial energy use (2,335.5 PJ in 2022) grew substantially, with mining energy use +23.9 percent and manufacturing coal use +25.9 percent year-on-year in 2022, reversing efficiency trends from earlier years (Tables 13-16, Statistics South Africa (2025)).

Import dependency and oil exposure. Despite coal self-sufficiency, South Africa's overall energy import dependency rose from 17.5 to 19.9 percent between 2015 and 2022 (Table 12, Statistics South Africa (2025)). Oil import dependency reached 93.1 percent following the closure of domestic refineries, and natural gas dependency is 99.9 percent (almost entirely from Mozambique). Oil products account for 84.8 percent of imports. The refinery-driven shift means South Africa now imports both crude and refined petroleum at global prices with no domestic buffer, a structural vulnerability the energy accounts quantify explicitly. Transportation consumes 433 PJ of oil products (2022), making it the largest end-use sector for liquid fuels (Table 19, Statistics South Africa (2025)).

Households and biomass. Household energy demand was 727.8 PJ in 2022 (12.0 GJ per capita), with biofuels (primarily fuelwood and charcoal) accounting for 46.5 percent of household energy, which is a figure comparable to Uganda's urban bioenergy share, indicating substantial energy poverty coexisting with a modern industrial economy (Tables 17-18, Statistics South Africa (2025)). Mining energy intensity reached 1.21 TJ per R million of GVA in 2022, the highest intensity among all sectors, highlighting the resource-intensive nature of the economy's dominant export industry.

4.8.3 Physical Asset Accounts for Land (SEEA-CF)

South Africa's land accounts, published in December 2020, cover the national mainland (121,966,453 ha) for 1990-2014 (Statistics South Africa, 2020). The national-level picture shows that natural and semi-natural land increased marginally (+826,384 ha, +0.8 percent), while cultivated land contracted (−347,267 ha, −2.1 percent), built-up land expanded (+196,735 ha, +6.5 percent), and waterbodies showed a large apparent decline of −675,852 ha or −32.2 percent, partly attributable to inter-annual variation in surface water extent between 1990 and 2014 wet and dry years, respectively (Table 2, Statistics South Africa (2020)).

Agricultural land dynamics. At Tier 2 detail, commercial crops declined by 411,696 ha (−3.5 percent) as rainfed commercial fields contracted sharply (−9.2 percent), while commercially irrigated pivots expanded by 221.4 percent, reflecting a structural shift toward high-value, water-intensive irrigated agriculture (Table 4, Statistics South Africa (2020)). Orchards and vines expanded +17.4 percent (+78,935 ha) and sugarcane fields grew alongside subsistence crops (+21,023 ha, +1.1 percent). Timber plantations contracted slightly (−35,529 ha, −1.9

percent), but informal settlements approximately doubled in extent (+95.7 percent, from approximately 31,000 to 60,000 ha), and mine footprints grew +15.9 percent (+42,981 ha; Table 4, Statistics South Africa (2020)).

Regional patterns. KwaZulu-Natal is the only province with a net loss of natural area (−178,241 ha, −2.8 percent), driven by simultaneous pressures from agriculture, urban expansion, and timber plantations, particularly in Harry Gwala district (−7.7 percent natural; Tables 8 and 11, Statistics South Africa (2020)). Nkangala district gained 30,274 ha of mining land, and Bojanala gained 7,511 ha, reflecting the platinum belt expansion north-west of Johannesburg. Pivot irrigation expanded most strongly in the Free State (+489.7 percent), underscoring the growing water demand from commercial agriculture in an already water-scarce interior province.

Ecosystem integrity. The land accounts integrate an Ecological Ensemble Index (EEI) to assess the condition of ecosystem types within biomes. Of 458 nationally defined ecosystem types, approximately 52 percent have EEI above 90 percent (near-pristine), but 18 percent have EEI at or below 60 percent, while six types have EEI at or below 20 percent (Tables 17-19, Statistics South Africa (2020)). Grassland has been most severely transformed, where historical extent of 33.1 million ha was reduced to 22.0 million ha by 2014; a loss of 11.1 million ha before the accounts begin, with Fynbos having 75 types below the 60 percent threshold (Figure 21, Statistics South Africa (2020)). Of all terrestrial ecosystem types, 22 percent are classified as threatened (NBA 2018, cited in Statistics South Africa (2020)).

4.8.4 Ecosystem Extent Accounts and Physical Ecosystem Services Accounts (SEEA-EA)

Biome extent. The terrestrial ecosystem extent accounts confirm that South Africa's dominant biomes are Savanna (39.4 million ha), Grassland (33.1 million ha), Nama-Karoo (24.9 million ha), and Fynbos (8.2 million ha), which together account for 87 percent of the country. Natural cover within biomes remained broadly stable between 1990 and 2014, showing that Savanna natural extent increased +0.8 percent, Grassland +1.2 percent, and Fynbos +0.8 percent, while Succulent Karoo was nearly unchanged (+0.06 percent; biome accounts Tier 1, Statistics South Africa (2020)). The Indian Ocean Coastal Belt, the most transformed biome, had only 551,628 ha natural in 1990 (47.2 percent) and remains at 48.1 percent EEI.

Biodiversity-based tourism (monetary ecosystem services). South Africa's experimental biodiversity tourism accounts, covering 2013-2019, provide the most detailed monetary ecosystem services estimates in the AFE region (Statistics South Africa, 2024). Total biodiversity-based tourism (domestic plus inbound) generated R43,389 million in internal spending in 2013, rising to R60,663 million in 2019. This represents approximately between 0.4 and 0.5 percent of GDP in biodiversity-specific tourism value added (BTDGDP: R19,622 million in 2013 rising to R27,726 million in 2019), which is 15.4 percent of total tourism GDP, declining to 13.3 percent by 2019 as general tourism grew faster than the nature-based segment (Table 1, Figure 1, Statistics South Africa (2024)). Total tourism employment reached

780,096 jobs in 2019 (4.8 percent of employment), of which 91,836 jobs (11.8 percent of tourism employment) were directly attributable to biodiversity-based activities; a scale of employment tied to ecosystems that no other country in the AFE region has quantified (Tables 22-23, Statistics South Africa (2024)). Avitourism alone was estimated to generate R2.24 billion per year (2010 DTIC estimate cited in Statistics South Africa (2024)).

Protected areas expansion. The protected areas accounts document one of the most sustained protected area expansion programmes on the continent (Statistics South Africa, 2021). Mainland protected area grew from 7.3 percent in 2000 to 9.2 percent in 2020 (11,280,684 ha), with the period 2015-2020 adding 1,182,456 ha, which is the largest five-year expansion on record. Marine protected areas expanded from 4,748 km^2 to 57,943 km^2 in 2019 (5.4 percent of the mainland Exclusive Economic Zone–EEZ–). By type, National Parks account for 37.4 percent of protected area (4,218,907 ha) and Nature Reserves 44.5 percent (5,022,911 ha; Table 4, Statistics South Africa (2021)). Biome coverage remains uneven, where Fynbos and Forest are better protected than Grassland and Savanna.

Estuarine ecosystem accounts. CSIR's experimental estuary accounts (Niekerk et al., 2020) cover 290 estuarine systems (total estuarine functional zone 200,730 ha), with condition tracked from approximately 1750 to 2018. The national estuarine condition index is 63.7 out of 100, but by area only 23 percent is in natural or near-natural condition while 63 percent is heavily modified (Tables 5.2-5.3, Niekerk et al. (2020)). The accounts report that approximately 12,000 million m^3 per year of freshwater is abstracted upstream of estuaries, with only 67 percent of natural flows now reaching estuarine systems (Table 6.1, Niekerk et al. (2020)). Key physical ecosystem services findings include net blue carbon sequestration potential lost since 1750 is approximately 1.37 million tonnes (Table 7.1, Niekerk et al. (2020)); fish nursery function is severely degraded, where dusky kob spawner biomass per recruit (SBPR) is below 2 percent of pristine, and 70-94 percent of key nursery habitats are under high or very high fishing pressure (Table 7.2, Niekerk et al. (2020)); and sawfish are now extinct in South Africa, with genetic exchange declining 15-21 percent for listed endemic species (Table 7.3, Niekerk et al. (2020)). The accounts flag that fish condition is "good" in only 4 percent of assessed estuaries, with alien fish present in 25 percent, underscoring the dual pressures of flow reduction and biological degradation.

4.9 Zambia (ZMB)

Zambia has compiled five SEEA accounts, all presented in two documents; a *Natural Capital Asset Compendium* covering land, forest, water, wildlife, and protected areas (Chisanga et al., 2026), and a standalone *Ecosystem Service Account* (Chisanga, 2026). Both were produced by the Ministry of Finance and National Planning with World Bank support under the Global Program on Sustainability (GPS). The government notes that approximately 40 percent of Zambia's national income derives from natural resources, motivating the full integration of natural capital accounts into the Eighth National Development Plan. The accounting period covers 2018-2023, a window defined by major floods and culminating in the catastrophic 2023/24

El Niño drought, which was declared a national disaster, affected more than one million farming households, and caused near-total failure of the staple maize crop.

4.9.1 Physical Asset Accounts for Land (SEEA-CF)

Zambia's land accounts, covering its national mapped area of 75.1 million ha, document rapid land-cover transformation between 2018 and 2023 (Table 2.1, Chisanga et al. (2026)).

Land cover changes. Tree cover (forests and woodlands) declined from 41,580,088 ha (55.4 percent) to 39,698,164 ha (52.8 percent), a loss of 1,881,924 ha over five years, equivalent to 376,385 ha per year (Table 2.4, Chisanga et al. (2026)). Cropland expanded from 2,603,203 ha (3.5 percent) to 3,563,653 ha (4.7 percent), a gain of 960,450 ha (+36.9 percent), while built-up area grew from 333,823 to 467,108 ha (+39.9 percent). Flooded vegetation (wetlands and dambos) contracted by 96,390 ha (−9.8 percent). Rangeland expanded by 894,672 ha (+3.2 percent). The natural land share (trees + flooded vegetation + rangeland) fell from 94.2 percent to 92.8 percent of national territory over just five years (Table 2.4, Chisanga et al. (2026)).

Transition flows. The dominant land conversion was trees to rangeland (4,921,342 ha, 44.7 percent of all conversions), reflecting the degradation of miombo woodlands to degraded grazing land rather than direct agricultural conversion (Table 2.3, Chisanga et al. (2026)). Rangeland-to-trees regeneration (3,259,379 ha, 29.6 percent) indicates significant natural regrowth, but net degradation still dominates. Rangeland-to-crops (1,214,202 ha, 11.0 percent) and trees-to-crops (251,019 ha, 2.3 percent) reflect expanding agriculture. At national level, conversions from natural to anthropogenic land (NACI = 2.13 percent) substantially exceed the area being restored or retained (RRI = 0.69 percent), yielding a net ecosystem degradation index (NEDI) of 1.44 percent over the accounting period (Table 4, Chisanga et al. (2026)).

Provincial patterns. North-Western Province is simultaneously the most stable (93.3 percent stability index) and most natural (99.3 percent natural land in 2023), with negligible degradation (NEDI = 0.12 percent). Eastern Province is the least stable (78.9 percent) and most rapidly transforming, with a NEDI of 5.25 percent driven by conversion of rangeland and woodland to crops, with Rangeland-to-crops (384,082 ha) and Trees-to-crops (47,586 ha) being the two largest transitions in the province (Table 4, Chisanga et al. (2026)). At district level, the highest tree losses were in Sesheke (Western, −300,094 ha), Kazungula (Southern, −188,375 ha), and Mumbwa (Central, −188,121 ha); while Mumbwa (+106,351 ha) and Kazungula (+75,443 ha) saw the largest cropland expansions (Tables 2.9-2.10, Chisanga et al. (2026)). The wetland loss index is highest in Southern Province (2.95 percent) and Lusaka (1.33 percent), while urban encroachment is most severe in Lusaka (0.75 percent; Table 5, Chisanga et al. (2026)).

4.9.2 Physical Asset Accounts for Timber Resources (SEEA-CF)

Zambia has produced three national forest account technical reports, covering 2010-2015, 2016-2021, and 2021-2024, under the WAVES and GPS programmes (Chapter 3, Chisanga et al. (2026)). The accounts compile physical supply and use tables (PSUTs) for wood forest products (WFPs) and non-wood forest products (NWFPs).

Charcoal and energy dependency. Charcoal and fuelwood together account for 78 percent of total energy consumed in Zambia (Chapter 3, Chisanga et al. (2026)), creating a direct structural linkage between the deforestation observed in the land accounts and household energy security. Unlicensed charcoal extraction vastly exceeds licensed production, representing a significant informal economy and a governance challenge for sustainable forest management. Indigenous poles supplied under licences declined by 310,303 m^3 between 2018 and 2023, while unlicensed indigenous poles fell by 2,507,891 m^3, which translates into reduced forest productivity. An estimated 60 percent of Zambians rely on medicinal plants, and NWFPs harvested include 109,968 tonnes of wild fruits, 83,177 tonnes of leaves, 74,454 tonnes of mushrooms, 44,842 tonnes of tubers, 2,097 tonnes of honey, and 1,477 tonnes of nuts per year (Figure 10, Chisanga et al. (2026)). The accounts note that the reduction in pole supply corresponds to declining net primary productivity and carbon storage.

4.9.3 Physical Supply and Use Tables for Water (SEEA-CF)

Zambia's water resources are organized around two major basin systems. On the one hand, the Zambezi Basin, with approximately 75 percent of national territory, including the Kafue and Luangwa sub-basins; and on the other one the Congo Basin, with approximately 25 percent of national territory, including the Chambeshi and Lake Tanganyika systems (Chapter 4, Chisanga et al. (2026)). Despite being considered water-rich by global standards, the water accounts highlight acute vulnerabilities in distribution, seasonality, and resilience.

Hydropower dependency and drought risk. The national economy is structurally exposed to water availability. Hydropower at Lake Kariba, Kafue Gorge, and Itezhi-Tezhi is the largest non-consumptive water user and underpins electricity supply nationwide. Lake Kariba levels fell below 15 percent of capacity in both 2016 and 2024, causing severe nationwide load-shedding and electricity imports. Agriculture is the largest consumptive user, with irrigation for wheat, sugarcane, and export crops competing directly with hydropower in the Kafue Basin. The Copperbelt's mining sector adds significant industrial water demand. The 2023/24 drought, which is the most severe in recent history, led to catastrophic maize crop failures in southern and central regions and forced massive electricity imports at substantial economic cost, illustrating the dual exposure of Zambia's economy to both water availability and hydropower generation (Chapter 4, Chisanga et al. (2026)). The water accounts acknowledge that a fully SEEA-Water-compliant national account has not yet been institutionalized, with monitoring networks remaining sparse and groundwater abstraction largely unmetered.

Wetlands and freshwater biodiversity. Wetlands cover approximately 19 percent of Zambia's total area, including globally significant complexes, like the Kafue Flats, Bangweulu Swamps, Lukanga Swamps, and the Barotse Floodplain (Chapter 4, Chisanga et al. (2026)). Dambos, seasonally waterlogged grassland depressions, cover 75,260 km^2, while swamps, marshes, and floodplains account for 30,104 km^2. These systems support approximately 490 freshwater fish species nationally, including 138 endemics in Lake Tanganyika alone. The Bangweulu wetlands alone generate an estimated US$11.7 million per year in ecosystem services (gross monetary value), with fishing and agriculture contributing 66.9 percent of total household cash income within the system (Chapter 5, Chisanga et al. (2026)). Water quality hotspots are documented in the Copperbelt (Kafue River, heavy metals from mining acid drainage) and Lusaka (pathogen contamination from septic systems and urban effluents contributing to cholera outbreaks).

4.9.4 Ecosystem Extent Accounts (SEEA-EA)

Using MODIS satellite data, Zambia's ecosystem extent accounts document biome-level changes between 2018 and 2023 (Table 8, Chisanga et al. (2026)). Deciduous vegetation, dominated by miombo woodlands, declined by 8.18 percent (607,452 to 557,791 km^2), consistent with the forest loss measured in the land accounts. Annual grass vegetation increased by 14.72 percent (236,766 to 271,613 km^2), reflecting woodland conversion to degraded grassland. Evergreen vegetation expanded by 35.59 percent (41,813 to 56,694 km^2), possibly reflecting the growth of managed and regenerating forests in wetter northern provinces. Urban and built-up land grew 3.42 percent in extent.

Carbon storage loss and monetary implications. The ecosystem accounts model total national carbon storage (using InVEST) at 18,732,820 million Mg C in 2018, declining to 18,110,965 million Mg C in 2023; a net loss of 621,854 million Mg C over five years (Section 5.6.2, Chisanga et al. (2026)). The Net Present Value of this carbon loss is estimated at USD 46 million (2018-2023), providing the only monetary ecosystem services estimate available for Zambia. Average annual Net Primary Productivity (NPP) declined by 150 kgC/m^2/year and GPP by 233 kgC/m^2/year over the same period, which can be interpreted from an ecological standpoint as reduced vegetation health linked to woodland degradation and drought stress.

4.9.5 Protected Areas Accounts (Thematic)

Zambia's protected areas and wildlife accounts, covering the period 2008-2015, were compiled under the WAVES programme and document wildlife-based provisioning services from Game Management Areas (GMAs; Chapter 5, Chisanga et al. (2026)). Bush meat supply from GMAs increased from 1,236.85 metric tonnes (2018) to 1,748.03 metric tonnes (2023), a 41 percent increase linked to growth in concession agreements and quota allocations. Revenue from bush meat grew from K45,807,799 (approximately US$3.85 million in 2018) to K118,413,018 (approximately US$4.60 million in 2023); a nominal increase of 158.5 percent, though partly reflecting the currency depreciation from ZMW 11.9/USD (2018) to ZMW 25.7/USD (2023; Figure 31, Chisanga et al. (2026)). Safari hunting accounts for between 70 and 80 percent of total bush meat supply and more than 90 percent of revenue. The accounts note that Zambia's national parks (South Luangwa, Kafue, and Lower Zambezi) anchor an eco-tourism industry that contributes to GDP and provides rural employment, but quantitative national totals for tourism revenue have not yet been compiled within the formal NCA framework.

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